Market Access and Urban Growth in the Former Soviet Union[*]

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Abstract

How does geopolitical fragmentation reshape urban growth? Using difference-in-differences on 1,396 cities over 1970–2021, I study the 1991 Soviet collapse. In post-Soviet Europe, annual population growth near new internal and old external borders fell by 0.4 and 0.6 percentage points relative to interior cities, implying populations 11–17% smaller after 30 years. Manufacturing exposure amplifies internal-border losses. Central Asian cities near previously sealed frontiers grew about 1 percentage point faster annually. The results suggest a market-access interpretation: fragmentation disrupts production links, and market transition exposes peripherality at old frontiers, which foreign access offsets only in Central Asia.

JEL classification: F15, N94, O18, P25, R11, R12

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1 Introduction

Geopolitical fragmentation has created new states from Eritrea to South Sudan, yet credible causal evidence on its effects on urban growth in developing countries remains scarce. Identification is difficult because border realignments reflect local political and economic conditions and coincide with other disruptions.

This paper exploits the 1991 collapse of the Soviet Union as a natural experiment to causally estimate how border realignments affect the economic geography of cities. The administrative boundaries that separated the Soviet republics were drawn between the 1920s and 1940s, primarily based on ethno-linguistic criteria and centralized administrative convenience, rather than the economic potential or growth trajectories of individual cities (Kramer and Smetana, 2013). For seven decades, these lines imposed minimal friction on trade and migration (Eichengreen, 1993; Suesse, 2018). The dissolution was driven by aggregate, macro-level crises such as the collapse of global oil prices, union-wide fiscal failures, and political upheaval in Moscow, rather than the localized economic conditions of individual cities (Gaidar, 2010). The collapse caught even seasoned scholars off guard (Fukuyama, 1992; Smith, 1993; Kotkin, 2001). Because the location of these borders was historically predetermined and the timing of their enforcement was driven by macro-political forces, the resulting spatial shock to market access is plausibly exogenous to the pre-existing economic trajectories of individual cities.

Existing studies examine episodes in which market access moves in the same direction for all treated locations at a given moment, whether the shock is division or integration (Redding and Sturm, 2008; Ahlfeldt et al., 2015; Brülhart et al., 2018). German division lowered market access for border cities and reunification later raised it, but the two shocks arrive in sequence, so the aggregate environment differs across them. The Soviet collapse instead hardened some frontiers and opened others simultaneously, so the sign of the border shock varies across cities while union-wide transition shocks do not.

In post-Soviet Europe, newly enforced internal borders severed the dense industrial supply chains that Gosplan had constructed across republican boundaries, while the simultaneous collapse of COMECON[1] redirected former Eastern Bloc partners toward Western Europe. In Central Asia, the dissolution opened previously sealed frontiers with Afghanistan, China, and Iran, borders that had been closed or heavily militarized throughout the Cold War (Pomfret, 2010, 2021). While the Central Asian border sample is smaller and geographically concentrated, suggestive evidence indicates these cities grew rather than shrank. [2]

Figure 1 maps the sample, the two border types, each city's exposure, and the two regional groupings used throughout. I estimate difference-in-differences models comparing cities within 75 km of borders to interior cities, using population data for 1,221 cities in post-Soviet Europe and

[1]The Council for Mutual Economic Assistance (COMECON) was an economic organization comprising the Soviet Union and its socialist allies.

[2]Section 3 defines market access as the composite of demand access and supplier access, the two dimensions of proximity to markets that jointly determine equilibrium city population. I do not construct a structural index (e.g., Donaldson and Hornbeck, 2016). Such an index is impractical here because Gosplan's centralized material balance system violates the competitive market assumptions underlying standard gravity frameworks (Eaton and Kortum, 2002), and parameterizing the heterogeneous trade costs of newly hardened post-1991 borders is not feasible.

175 in Central Asia spanning 1970–2021. The long pre-period permits direct testing of parallel trends, and the three-decade post-collapse horizon captures persistent effects. I distinguish two border types: *internal borders*, the former inter-republican boundaries that became international after 1991, and *external borders*, the frontiers that were already international throughout the Soviet period. Newly international borders raised trade costs. At existing external frontiers, market transition could expose locational disadvantages previously buffered by administrative allocation, even as crossing became easier. The net effect depended on whether improved foreign access offset the loss of that insulation.

The main finding is that annualized population growth in post-Soviet European cities near internal and external borders fell by about 0.4 and 0.6 percentage points (pp), respectively, relative to interior cities. Sustained over three decades, these differences imply relative population shortfalls of roughly 11–17%. The exposure decomposition shows declines among internal-only cities and particularly large losses among cities near both frontiers; the external-only estimate is also negative, although less precisely estimated. In Central Asia, cities near external borders grew about 1 pp faster than interior cities, while the full-sample internal-border estimate is not distinguishable from zero. Formal interaction tests establish that the border coefficients differ across the two regions. I interpret the internal-border decline as lost market access from severed cross-republican supply links, the European external-border decline as a penalty for peripheral location once markets replaced planning, and the Central Asian gain as likely improved access to newly opened markets (Pomfret, 2021).

Three pieces of evidence are consistent with this interpretation. First, internal border effects in Europe and the external gain in Central Asia attenuate with distance from the frontier, and Europe's external decline is concentrated among inland cities: it is significantly smaller, and not detected, among the 23 coastal cities, where a port offers an outlet the land frontier does not close. Second, I exploit sectoral heterogeneity in exposure to cross-border linkages: manufacturing-intensive cities, more dependent on the vertically integrated supply chains that spanned republican boundaries, experienced amplified losses where borders hardened in post-Soviet Europe. Third, the results survive nearest-neighbour and exact matching on pre-1991 city size, industrial employment and composition, and defense sector presence, indicating that the estimates do not reflect pre-existing differences in observable characteristics. The timing also tells against the post-Soviet resettlement wave: return migration peaked in the mid-1990s, whereas the border effects are largest in the following decade and persist into the 2010s.

This paper contributes to the new economic geography literature, which since Krugman (1991) has emphasized market access as a central determinant of the spatial distribution of economic activity (Helpman, 1998; Fujita et al., 2001; Redding and Venables, 2004). Empirical work has tested this prediction through border or administrative frictions (Behrens, 2024; Egger et al., 2022; Brülhart et al., 2018; Ahlfeldt et al., 2015; Redding and Sturm, 2008), transport infrastructure (Jedwab and Storeygard, 2022; Büchel and Kyburz, 2020; Baum-Snow et al., 2017; Storeygard, 2016; Faber, 2014), and trade policy (Dix-Carneiro and Kovak, 2017; Autor et al., 2013; Hanson, 2001). Closest to this paper, Suesse (2018) uses a gravity framework to show that Soviet disintegration severed inter-republican supply chains as nationalist mobilization

advanced and documents the trade collapse at the republic level; I trace its spatial incidence across individual cities, so the population responses estimated here complement the trade-based results established at the aggregate.

This paper makes two contributions. First, it exploits a geopolitical rupture that simultaneously hardened some frontiers and opened others, so the sign of the border shock varies across cities within a single episode rather than across episodes, and it does so at a scale, nearly 1,400 cities across 15 countries, that permits tests of distance gradients and sectoral exposure. Second, it provides city-level evidence that the economic consequences of an existing frontier can change with the allocation regime. Cities along post-Soviet Europe's old external frontier declined relative to interior cities although these boundaries were not newly imposed and became relatively easier to cross. That market transition exposed the locational disadvantages of Soviet industrial siting is itself familiar (Hill and Gaddy, 2003); what is new is the evidence on international frontiers specifically, on the population trajectories of the cities beside them, and on the role of alternative access.

More broadly, the findings give city-level evidence on the costs of political fragmentation theorized by Alesina and Spolaore (1997, 2005), and on the persistence of city sizes after major disruptions (Miguel and Roland, 2011; Brakman et al., 2004; Davis and Weinstein, 2002): border realignments generate lasting spatial divergence where they durably rewire access to nearby markets.

2 Institutional Background

The dissolution of the Soviet Union in 1991 fragmented one of the most deeply integrated economies in the modern world. Since its founding in 1922, the USSR organized a vast and diverse population into 15 Soviet Socialist Republics through national-territorial delimitation (*razmezhevanie*). Administrative borders between republics imposed little friction on internal trade or migration (Eichengreen, 1993; Denisenko and Chudinovskikh, 2017). Republican agencies retained limited planning autonomy and could on occasion influence allocation within their territory, but their remit extended to the republic as a whole rather than to particular locations within it. Gosplan, the State Planning Commission, coordinated production through hierarchical negotiations, deliberately constructing vertically integrated industries that spanned republican boundaries. Djankov and Freund (2002) estimate a gravity equation for trade among Russian regions and FSU republics and find that Russian regions traded no more with one another than with other republics before the collapse. A pronounced domestic bias emerged afterward as inter-republican trade contracted, with Russian regions trading roughly 60% more among themselves than with the republics.

Spatial frictions in the Soviet economy were organized around individual cities rather than administrative lines: the *propiska* residency permit regime, introduced in 1932, restricted access to high-status cities such as Moscow, Saint Petersburg, and Tashkent rather than movement across republican borders (Buckley, 1995; Light, 2012; Gang and Stuart, 1999).

2.1 Soviet Europe

The European republics contained the Union's most industrialized economies, far more densely integrated through planned supply chains than Central Asia (Pomfret, 2021), with factories often relying on single suppliers in other republics for critical inputs. Blanchard and Kremer (1997) demonstrate that when central planning's enforcement mechanism disappeared before market institutions emerged, bargaining broke down across these supply chains. Output fell farthest for goods with the most complex production processes, precisely because they depended on cross-republican coordination.

On the western frontier, Soviet external trade with Eastern Europe was organized through COMECON, which accounted for roughly half of Soviet foreign trade (Holzman, 1985).[3] Foreign trade as a whole was nonetheless small next to trade among the republics, which absorbed more than 80% of the exports of every republic except Russia in 1990 (Al-Atrash and Havrylyshyn, 1998; Djankov and Freund, 2002).

COMECON effectively dissolved on January 1, 1991, when members shifted to hard currency transactions. Intra-bloc trade dropped from 13 times gravity-predicted levels in 1990 to just 4 times by 1993 (Maurel and Cheikbossian, 1998). The western frontier also became harder to cross as the Central European states prepared for EU accession. Residents of Belarus, Moldova, Russia, and Ukraine could enter Poland without visas in the 1990s, but Poland introduced visas in October 2003, Romania required them of Moldovans on its accession in January 2007, and costlier Schengen visas followed in December 2007.

2.2 Central Asia

Central Asian republics were integrated with Russia mainly through extractive north-south corridors for cotton, natural gas, and minerals, and lacked the dense inter-republican linkages characteristic of the European USSR (Pomfret, 2021), while over 80% of Soviet industrial output was concentrated in Russia and the western republics (IMF et al., 1991). All roads, railways, and pipelines led north to Russia.

Before 1991, Central Asian republics were nearly isolated from neighboring non-Soviet states. The Sino-Soviet border was closed in 1962 and remained heavily militarized through the Cold War (Radchenko, 2009), contact across the Iran-Turkmenistan border was channeled through Moscow, and the Afghanistan border became a war zone after the 1979 Soviet invasion; the first Central Asian rail link not routed through Russia opened only in 1990 (Pomfret, 2021). The dissolution opened all three frontiers simultaneously. Shuttle traders (*chelnoki*) built cross-border bazaar economies at newly accessible crossings (Spector, 2017), Iran and Turkmenistan began direct bilateral trade, and in 1992, when the Khorgos crossing on the Kazakhstan-China border reopened, approximately 700,000 Kazakhstani citizens crossed to China for "shop tourism" (Peyrouse, 2008).

[3]Trade with Türkiye and Iran remained minimal throughout the Cold War. The Armenian-Turkish border stayed closed after 1991 due to the Nagorno-Karabakh conflict. Türkiye and Azerbaijan share only a 17 km land border, offering little commercial potential in the early post-Soviet years, and Georgia's early-1990s wars further disrupted the frontier.

Appendix Figure A1 documents the initial reorientation: from 1991 the share of non-FSU trade conducted with former Eastern Bloc partners fell in the European republics and Russia, the share conducted with China and Afghanistan rose in the Central Asian republics, and interrepublic trade contracted to roughly one-fifth of its 1990 level by 1994, when the Soviet-format series ends.

The nature of the market access gained in Central Asia also differed qualitatively. Central Asian border cities gained access to lower-income neighbors: Afghanistan, Iran under sanctions, and China's least developed western provinces. As Kaminski and Mitra (2012) and Karrar (2019) document, the resulting cross-border trade was dominated by cheap consumer goods for resale in domestic bazaar economies, requiring little fixed investment and allowing gains to materialize rapidly through informal shuttle trade.

The Soviet collapse also triggered significant migration, particularly among ethnic minorities. Roughly a million ethnic Germans and Jews emigrated to Germany and Israel, while ethnic Russians returned to Russia from other republics (Becker et al., 2012; Heleniak, 2004). The return flow was front-loaded: net migration into Russia peaked at over 800,000 in 1994 and by 2001 had fallen below a tenth of that level (Heleniak, 2004). These flows were predominantly repatriations to ancestral homelands rather than localized sorting away from borders, and the largest emigrant populations were concentrated in major interior cities (Mukhina, 2007; Heleniak, 2004). Section 5 tests whether such resettlement could account for the border effects.

3 Conceptual Framework

This section sets out the spatial equilibrium logic that organizes the analysis. I draw on the economic geography models of Krugman (1991), Helpman (1998), and Fujita et al. (2001), in the multi-region form used by Redding and Sturm (2008) and extended to intermediate inputs following Redding and Venables (2004). I use the framework to sign and interpret reduced-form estimates rather than to estimate structural parameters.

The framework describes adjustment under market allocation. Before 1991, Gosplan coordinated production and distribution, while the *propiska* system regulated mobility by city status. Administrative allocation could buffer locational disadvantages by sustaining links to distant suppliers and buyers. Market transition could therefore make access to nearby markets more consequential as some frontiers hardened and others opened. I assess pre-transition differences in growth empirically in Figures 2 and 3.

Workers are mobile and consume traded varieties and a local amenity in fixed supply. Firms produce differentiated varieties under monopolistic competition using labor and a composite intermediate, and both final and intermediate goods bear iceberg trade costs $\tau_{ic} > 1$. The price index of the intermediate bundle is the same trade-cost-discounted aggregate that governs the consumer cost of living, so a city's marginal cost is decreasing in its access to suppliers. Two distinct objects therefore summarize a city's position in this network. Supplier access is the trade-cost-discounted measure of varieties a city can obtain as inputs,

$$SA_c = \sum_i n_i \left(p_i \tau_{ic}\right)^{1-\sigma}, \tag{1}$$

and demand access is the trade-cost-discounted measure of expenditure its firms can reach,

$$DA_c = \sum_i \left(w_i L_i\right) P_i^{\sigma-1} \tau_{ci}^{1-\sigma}, \tag{2}$$

where n_i is the mass of varieties produced in i and P_i its price index. Spatial equilibrium requires indirect utility to be equalized across populated cities, which yields equilibrium population as

$$L_c = \kappa H_c SA_c^{\gamma} DA_c^{\eta}, \qquad \gamma, \eta > 0, \tag{3}$$

where H_c is the local amenity stock and γ and η depend on σ, the expenditure share on traded goods, and the intermediate share in production.

Supplier and demand access both depend on bilateral trade costs, so frontier hardening reduces both. Population responses cannot separately identify these margins; I call their composite *market access*. Supplier access is particularly relevant in the Soviet setting because input deliveries were difficult to replace (Blanchard and Kremer, 1997).

Within the market-allocation regime, the population response to a change in τ depends on the direction of that change and the share of a city's input purchases and output sales crossing the affected frontier. Where enforcement raised trade costs on a frontier carrying a large share of a city's linkages, equation 3 predicts decline; where a previously prohibitive frontier opened, it predicts growth. Opposite-signed regional effects therefore follow from one comparative static rather than separate theories, and the regional difference is a share parameter rather than a difference in institutions or planning regime. I test this with interactions for Central Asia.

At existing external frontiers, changes in crossing costs coincided with the transition away from administrative allocation. Planning could sustain links to distant suppliers and buyers, buffering the disadvantages of a truncated hinterland. Market transition could therefore increase the cost of peripheral location even where crossing became easier. Because equations 1 and 2 discount economic mass rather than realized trade, this argument does not require frontier cities to have conducted a disproportionate share of Soviet foreign trade. The net population response depends on whether improved foreign access offsets the loss of this insulation. In Europe, the contraction and westward reorientation of neighboring economies could limit that offset, while subsequent EU visa requirements could impede recovery (Section 2.1). Redding and Sturm (2008) likewise find smaller effects from German reunification than from division. This interpretation requires a change in the allocation regime as well as in border frictions. Equations 1 to 3 describe the market equilibrium, within which easier foreign access, holding other conditions fixed, raises both SA_c and DA_c. The design does not identify the change in allocation regime separately from fragmentation itself.

In Central Asia the comparative static runs through the other margin. A lower share of production there ran through inputs sourced across republican boundaries, so enforcing internal frontiers severed fewer supplier links and the framework predicts a smaller internal effect, though

individual republics departed from the regional average. What the opened external frontiers supplied was buyers, in the bazaar economies documented by Kaminski and Mitra (2012), and buyers enter DA_c rather than SA_c. The European internal-border decline is therefore primarily a loss of supplier access and the Central Asian external-border gain primarily an addition of demand access. What determines exposure is not sector as such but the share of a city's purchases and sales crossing the frontier whose τ changed. Manufacturing employment tracks that share in Europe but not in Central Asia, where manufacturing faced north toward Russia rather than toward the frontiers that opened, so low agricultural employment proxies instead for orientation toward cross-border commerce.

Beyond the sign, the framework predicts that effects should be localized, decaying with distance from the changed frontier, since cities farther away already faced high trade costs to locations across it. What decays is the hinterland a city forgoes, which depends on the geometry of the frontier and on city size, so the decline need not be strictly monotone in straight-line distance. The sea enters the same accounting. A coastline truncates a city's overland hinterland as a closed frontier does, but a port reaches distant foreign mass at low cost, so a land frontier accounts for a smaller share of a coastal city's access and should cost it less. Effects should also be larger for small cities, since own-city expenditure enters DA_c at $\tau_{cc} = 1$ and is therefore a larger share of a large city's total access, so a given loss of external mass is proportionally smaller.

In the framework, population adjusts through internal migration: a loss of market access lowers real wages, inducing workers to leave until utility is equalized across cities. Observing migration does not, however, identify the shock that induced it. Ethnic resettlement, changes in mobility restrictions, and other local shocks correlated with border proximity could also generate differential population growth. The two accounts also differ in timing. Resettlement was concentrated in the first half of the 1990s, so a border effect produced by it should appear in the first post-collapse census period and fade afterwards, whereas a durable loss of market access predicts relative decline that persists. The timing, size, coastal-access, and sectoral comparisons in Section 5 assess whether the observed patterns support a market-access interpretation, but do not separately identify migration motives.

4 Data and Empirical Methodology

The empirical analysis draws on the following data sources:

1. Population census. The panel follows 1,396 former Soviet cities with over 5,000 residents in 1970 through 2021.[4] Population and coordinates come from Brinkhoff (2021), supplemented by Demoscope Weekly, the Russian Institute of Demography's database.[5] Observations cover the 1970, 1979, and 1989 Soviet censuses and three post-1991 periods, with dates varying by country (Table A1).

[4] The results are robust to using a threshold of 10,000.

[5] Available at citypopulation.de and demoscope.ru, respectively.

I use the annualized growth rate of population following Redding and Sturm (2008), as it allows comparing growth rates consistently across different census periods, adjusting for length variations in those periods.[6]

For each city I also compute straight-line distances to the nearest internal border, the nearest external border, and the coast. External border distance is measured to the nearest non-FSU territory, which for all but three external border cities lies across a land frontier; Tallinn, Maardu, and Paldiski fall within 75 km of Finland across the Gulf of Finland. Coastal cities are those within 75 km of the open sea, excluding the landlocked Caspian.

2. Industry data from 1989. The paper uses total industrial employment, turnover, manufacturing output in the civilian industry, and the SIC code of each city's primary sub-industry from 1989, compiled by Kofanov and Mikhailova (2015) from Soviet statistical yearbooks. Matching their Cyrillic city names to the main sample, described in Appendix A1, recovers 1,225 of the 1,396 cities (1,076 of 1,221 in Europe and 149 of 175 in Central Asia); the unmatched are predominantly small places renamed after independence or without reported sectoral employment.

3. Defense industry data. Data on Soviet defense enterprises comes from Dexter and Rodionov (2024), who compiled a comprehensive database of military factories and research establishments across the USSR. I restrict the sample to enterprises whose military work ended between 1989 and 1991 and whose workforce size is documented, and construct for each matched city the count of such enterprises and the sum of their ordinal workforce sizes. Appendix A1 describes the filtering and name-matching.

Because sub-national GDP is unavailable across the FSU, Appendix C provides some corroborative results with nighttime lights from the Defense Meteorological Satellite Program (DMSP), 1992–2013, as a proxy for city-level economic activity (Henderson et al., 2012); Appendix A1 describes the cleaning. Finally, Appendix Tables A2 and A3 report descriptive statistics by region.

Empirical Strategy

The identifying variation is the differential change in population growth of border relative to interior cities after versus before the dissolution, pooled across post-Soviet countries that operated under the same centralized institutional framework before 1991. What identification requires is weaker than the absence of friction. Whatever autonomy republican agencies exercised operated at the level of the republic and applied alike to its interior and its border cities, so it gives no reason for cities near administrative boundaries to follow different growth trajectories from interior cities in the same country.

[6] It can be computed in city c as

$$y_{ct} = \frac{ln(P_{ct}) - ln(P_{c,t-L_{ct}})}{L_{ct}}, \tag{4}$$

where P_{ct} is the population statistic at time t, and L_{ct} is the number of years between t and the prior census. Log differences are symmetric, additive across periods, and nearly identical to the discrete compound annual growth rate at the typical growth rates in this sample (mean 0.68%).

Formally, the DiD approach does not require that border locations were randomly assigned but assumes that, absent the sudden dissolution, trends in the outcome variable would have been parallel between border and non-border cities (Roth et al., 2023; Meyer, 1995).[7]

The baseline specification is

$$y_{ct} = \alpha_c + d_t + \beta_1 \left(IB_c \times \text{Post}_t\right) + \beta_2 \left(EB_c \times \text{Post}_t\right) + \epsilon_{ct}, \tag{5}$$

where IB_c and EB_c equal one for cities within 75 km of an internal and an external frontier respectively, Post_t equals one for census periods after 1991, α_c and d_t are city and period fixed effects, and standard errors are clustered at the city level. The coefficients β_1 and β_2 give the change in annualized population growth for border relative to interior cities after dissolution. Column (2) of Table 1 adds a Central Asia indicator interacted with Post_t and with each treatment, so that β_1 and β_2 report effects for post-Soviet Europe and the triple interactions report the Central Asian differences.

To validate the parallel trends assumption and examine dynamic effects over time, I employ an event-study framework:

$$y_{ct} = \alpha_c + d_t + \sum_{k=T_0}^{-2} \delta_{1,k}\left[IB_c \times D_k\right] + \sum_{k=0}^{T_1} \delta_{1,k}\left[IB_c \times D_k\right] + \sum_{k=T_0}^{-2} \delta_{2,k}\left[EB_c \times D_k\right] + \sum_{k=0}^{T_1} \delta_{2,k}\left[EB_c \times D_k\right] + \epsilon_{ct} \tag{6}$$

where y_{ct} is the annualized growth rate of population in city c in census period t,[8] IB_c and EB_c are defined as in equation 5, and D_k is an indicator equal to one when the observation falls in event-time period k relative to the Soviet dissolution (with $k = 0$ denoting the first post-collapse census period).

The event window is $k \in [-2, +2]$, with $k = -1$ omitted: the coefficients at $k = -2$ provide a pre-treatment comparison, while those at $k \geq 0$ trace subsequent adjustment. Because census dates differ across successor states, event time indexes relative census periods rather than calendar years.

The baseline comparison follows the geography of the integrated pre-dissolution economy. Before 1991, cities on either side of republican boundaries operated within a common planning system, with production and distribution organized across those boundaries. Interior cities elsewhere in that system therefore provide an economically motivated comparison group for cities exposed to its fragmentation. City fixed effects absorb persistent local differences, period effects capture common shocks, and sub-region-by-period effects allow broader regional trajectories to differ in robustness specifications. Country-by-period effects would instead identify the coefficients solely from within-country variation, absorbing both national adjustment to dissolution and other national shocks. The baseline retains this cross-country variation under the identifying assumption that, absent dissolution, border and interior cities would have experienced parallel

[7] While spatial spillovers from border to interior cities technically violate SUTVA (Rubin, 1980), capturing this relative divergence is the goal of the DiD framework, consistent with spatial equilibrium models (Fujita et al., 2001; Crozet, 2004; Redding and Sturm, 2008).

[8] It covers five intercensal periods – 1970-1979, 1979-1989, 1989-1999, 1999-2009, and 2009-2019 – with variations after the collapse depending on specific country census years.

changes in growth. The common institutional history motivates this comparison, while the pre-treatment evidence assesses its plausibility.

5 Results

DiD Estimates

Table 1 begins with the pooled sample of all 1,396 cities. Column (1) imposes common border coefficients across regions and estimates declines in annualized population growth of 0.210 pp near internal borders and 0.563 pp near external borders. Column (2) allows the coefficients to differ between post-Soviet Europe and Central Asia. The European internal and external estimates are -0.397 and -0.591 pp, respectively, and a joint test rejects common border coefficients across regions ($p = 0.006$). The Central Asia-by-post coefficient is small and statistically insignificant (0.079 pp, $p = 0.807$), providing no evidence of a differential post-1991 change among interior cities.

The implied Central Asian coefficients in Column (2) are 0.411 pp for internal borders ($p = 0.256$) and 0.781 pp for external borders ($p = 0.074$). Columns (3)–(4) estimate the regions separately, allowing their period effects to differ. The European estimates are -0.402 and -0.595 pp, both significant at the 1% level. In Central Asia, the internal estimate is 0.471 pp and statistically insignificant, while the external estimate is 0.969 pp ($p = 0.033$). Sustained over three decades, the latter growth difference implies a relative population increase of roughly one-third. The Central Asian external-border sample contains 15 cities: 12 along the Afghanistan frontier, two near the Iran–Turkmenistan border, and one near the Kazakhstan–China border. Its small size and geographic concentration warrant caution.

Columns (5)–(6) distinguish single from overlapping exposure. In Europe, 80 cities lie near both frontiers, accounting for 56% of external-border cities and 23% of internal-border cities. Internal-only cities experience an estimated growth decline of 0.358 pp ($p = 0.001$), while dual-exposure cities decline by 1.099 pp ($p < 0.01$). The external-only estimate is also negative, at -0.441 pp, but is less precise (standard error 0.291). Among cities near an internal frontier, external exposure is associated with an additional decline of 0.740 pp ($p = 0.007$). Although the dual-exposure decline exceeds the sum of the two single-exposure declines in magnitude, the test does not reject additivity ($p = 0.456$).

In Central Asia, the external-only estimate is 1.356 pp and significant at the 5% level, showing that the positive external-border result is not confined to cities near both frontiers. The internal-only estimate is 0.552 pp and statistically insignificant. Only five cities have dual exposure, limiting the precision of comparisons involving that category. The table reports the number of cities in each exposure category for every estimation sample.

Event-Study Estimates

Figure 2 indexes total border and interior city population to 1970 and plots their difference, separately by border type and region. In all four panels border and interior cities move in parallel before 1991. Afterwards, European border cities decline sharply and persistently relative

to the interior, more so near external borders, while in Central Asia border cities grow faster in relative terms, most visibly near external borders.

Figure 3 presents event-study estimates from equation 6, with $k = -1$ as the omitted reference period. In Europe, the pre-collapse coefficient at $k = -2$ is close to zero for both border types, supporting the parallel trends assumption. In Central Asia, internal border pre-trends are similarly flat, while external border cities show a marginally significant negative coefficient at $k = -2$, which may reflect noise in a small sample but warrants caution in interpreting the Central Asian external border results.

In Europe, both border types show sharp and persistent declines after dissolution, with effects peaking at $k = 1$ and remaining strongly negative through $k = 2$. For external borders, the persistence through $k = 2$ is consistent with the westward reorientation of the economies across the frontier. For internal borders, the delayed peak is consistent with Blanchard and Kremer (1997)'s account of cascading supply chain failures. The persistence of effects three decades after dissolution indicates lasting spatial divergence rather than temporary disruption.

The timing also argues against the post-Soviet resettlement wave as the source of these effects. Return migration to Russia peaked in the early 1990s, so it falls mostly within $k = 0$. The border effects are instead largest at $k = 1$ and remain negative at $k = 2$, and dropping the 1989–1999 period does not remove them (Table A7, Column 2).

In Central Asia, the pattern reverses. Internal border effects remain small and statistically insignificant across all post-dissolution periods, consistent with limited inter-republican trade to disrupt (Pomfret, 2010). External border cities show a gradual emergence of positive effects. The coefficients at $k = 0$ and $k = 1$ are positive but imprecisely estimated, consistent with the initially informal and uneven nature of cross-border commerce as shuttle traders began exploiting newly opened frontiers (Kaminski and Mitra, 2012). By $k = 2$, the effect becomes significant at the 5% level.

Figure A2 confirms that the event-study patterns are robust to replacing period fixed effects with sub-region-by-period fixed effects in Europe and to excluding Turkmenistan in Central Asia. The external border effects in Europe are, if anything, larger and more precisely estimated under this more demanding specification.

Robustness

Table A4 examines robustness to sub-region-by-period fixed effects and to excluding Turkmenistan from the separate Central Asian specifications. The regional pattern is preserved. In the pooled common-coefficient model, the external estimate remains close to its baseline value at -0.540 pp ($p < 0.01$), while the internal estimate attenuates to -0.159 pp and is statistically insignificant. Allowing regional interactions again rejects common border coefficients ($p = 0.002$, Column 2). The European internal and external estimates remain negative and statistically significant at -0.241 pp ($p = 0.034$) and -0.663 pp ($p < 0.01$), respectively (Column 3). The internal magnitude is smaller than in the baseline, whereas the external magnitude is slightly larger.

The European decomposition in Column (5) likewise preserves the negative signs. The external-only and dual-exposure estimates, −0.437 and −1.047 pp, are close to their baseline values of −0.441 and −1.099 pp. The internal-only estimate attenuates to −0.172 pp and is statistically insignificant. The incremental external penalty among internal-border cities remains precisely estimated at −0.875 pp ($p = 0.001$), and additivity is again not rejected ($p = 0.279$).

Columns (4) and (6) exclude Turkmenistan, whose census timing differs from the other Central Asian countries. Because Central Asia constitutes a single sub-region, these specifications use time controls equivalent to those in the baseline regional regressions. The additive external estimate remains positive and similar in magnitude at 0.838 pp ($p = 0.055$); the external-only estimate is 1.192 pp ($p = 0.060$), compared with 1.356 pp in the full sample. The internal estimates increase to 0.714 pp in the additive specification ($p = 0.054$) and 0.778 pp in the decomposition ($p = 0.049$), indicating greater sensitivity to sample composition on this margin. The external-only category contains eight cities after the exclusion.

Table A5 splits the sample at the median 1970 city population, and the European effects are carried almost entirely by small cities (column 3). Below the median, internal border cities decline by 0.447 pp and external border cities by 0.587 pp, both significant at the 5% level. Above the median the estimates fall to negative 0.130 and 0.200 pp and neither is distinguishable from zero. The Central Asian external border gain follows the same pattern, with small cities growing 1.402 pp faster.

The larger point estimates for small cities are consistent with Section 3 and the pattern reported by Redding and Sturm (2008) for German division. This pattern is less readily explained by an account centered on departures from large urban centers, but city size alone cannot distinguish economic migration from ethnic resettlement or changes in mobility restrictions.

If market transition raised the cost of peripheral location, the external border penalty should be smaller where maritime access offers an alternative route to suppliers and customers. Table A6 uses each city's distance to the open-sea coastline. Allowing the 164 cities within 75 km of the coast their own post-1991 trajectory leaves the European estimates at −0.397 and −0.592 pp (Column 1), and dropping them yields −0.368 and −0.784 pp (Column 3). The estimates are essentially unchanged under sub-region-by-period fixed effects and under alternative coastal definitions (Columns 2, 8, and 9). Column 4 lets the border effects differ by sea access. Inland external border cities decline by 0.782 pp, while the 23 coastal ones, listed in Appendix A1, show no detectable decline (0.377 pp; $p = 0.007$ for equality). The gap does not rest on the three Estonian cities whose nearest foreign territory lies across the Gulf of Finland or on Kaliningrad, which Appendix B treats as a special case. Excluding the Estonian cities widens it ($p = 0.002$) and excluding Kaliningrad narrows it ($p = 0.051$), while excluding both leaves 16 coastal external border cities at 0.391 pp against −0.782 pp inland ($p = 0.019$; Columns 5–7). Internal border effects do not differ by coastal access (−0.383 and −0.366 pp, $p = 0.951$), consistent with a shock that runs through cross-republican supply links a port does not readily replace.

Table A7 reports additional sensitivity checks. The results are robust to excluding large cities subject to *propiska* restrictions, which also addresses ethnic composition since these cities hosted disproportionately diverse populations and experienced the highest outmigration rates

after 1991 (Heleniak, 2004) (Column 1); to dropping the turbulent 1989–1999 intercensal period, after which the European external border effect is -0.648 pp, so the external effect is not a product of the disruption and resettlement of the early 1990s (Column 2). Column (3) drops dual-exposure cities, providing a sample-exclusion counterpart to the decomposition in Table 1, Columns (5)–(6).

Column (4) excludes conflict-affected border pairs (e.g., Armenia–Azerbaijan and Georgia–Russia). Removing conflict zones *strengthens* the European internal border effect to -0.690 pp ($p < 0.01$), showing that the estimate does not depend on including those conflict-affected pairs. The external border effect remains marginally significant (-0.412 pp, $p = 0.071$).

Column (5) applies a region-specific restriction addressing how the regions are classified. In Europe it drops the 193 Russian cities in the Siberian, Far Eastern, and Urals federal districts; the internal border effect strengthens to -0.532 pp and the external effect holds at -0.534 pp, both significant at the 1% level, so the European estimates do not depend on grouping Asian Russia with post-Soviet Europe. In Central Asia the column drops the ten cities on the Kazakhstan–Russia frontier, the one Central Asian internal border that severed a dense industrial corridor. The internal estimate rises from 0.471 to 0.569 pp and remains insignificant.

Column (6) restricts the control group to cities within 750 km of any border. In Europe the internal negative effect strengthens to -0.529 pp while the external effect attenuates to -0.392 pp but remains significant at the 5% level, consistent with a peripherality component in the external estimate that the restriction partly removes (see the distance results below). In Central Asia neither effect moves much.

Allowing for spatial correlation across nearby cities, Conley standard errors at cutoffs from 25 to 200 km preserve all significance levels in both regions (Table A8; Conley, 1999; Colella et al., 2023). Figure A3 presents leave-one-out analyses. In Europe, both border effects remain negative across all ten country exclusions. The external estimate is smallest and least precise when Russia is dropped, which removes most of the interior control group. In Central Asia, the external border coefficient remains positive across all five exclusions, but both effects are sensitive to dropping Kazakhstan, which contributes the largest share of interior control cities and leaves too few for precise estimation.

Appendix Table A9 replaces the binary treatment with 25 km distance bins, following Redding and Sturm (2008). In Europe, internal border effects decay with distance, from a decline of 0.486 pp at 0–25 km (significant at 1%) to 0.363 pp at 25–50 km (5%), and are insignificant beyond 50 km. External border effects are largest at 0–25 km (-1.055 pp) but non-monotone beyond it: the 25–50 km coefficient is insignificant, while the 75–100 km coefficient (-0.835 pp) is nearly as large as the first. This irregular gradient is consistent with a broader peripherality component if market transition increased the importance of access to nearby markets. I therefore rely on internal borders for the distance-gradient evidence of a localized frontier effect. Under sub-region-by-period fixed effects (Column 2) the external gradient strengthens while internal border effects attenuate. In Central Asia, cities within 25 km of external borders grew 1.438 pp faster (significant at 5%), with positive but imprecise coefficients farther out, consistent with the localized shuttle trade documented by Kaminski and Mitra (2012).

Table A10 redefines the binary treatment at thresholds from 25 to 125 km. The European internal border effect attenuates monotonically, from -0.537 pp at 25 km to -0.178 pp at 125 km, as a frontier-specific shock implies, whereas the external effect increases from -1.051 to -0.595 pp and then declines to -0.833 pp, with the standard error more than halving, consistent with the same peripherality component. In Central Asia the external gain is significant at the 5% level at every threshold up to 100 km (15 treated cities at 75 km, 19 at 100 km), while the internal estimate is insignificant throughout.

The wider rings also address the Soviet 101st-kilometre restriction, which limited residence around major cities rather than near frontiers, and the results are robust to excluding the cities it surrounded (Table A7, Column 1).

Appendix B reports border-pair estimates: most European internal pairs are negative and all Central Asian external pairs positive, though pairs with few cities are imprecise. Appendix C reports nighttime lights from 1992, which reproduce the cross-regional contrast at external borders in the first post-collapse decade but otherwise corroborate the population results only partially.

Sectoral Heterogeneity

I test the sectoral prediction of Section 3 using 1989 industrial employment by sector (Kofanov and Mikhailova, 2015), decomposing the European border effects by manufacturing quartile and splitting Central Asian cities at the within-region median of agricultural employment.[9] Appendix Tables A12 and A13 report splits for the remaining sectors.

In Europe (Table 2, Columns 1–2), manufacturing shares are compressed (mean 0.834, median 0.925), so a linear interaction is hard to interpret and each quartile receives its own treatment effect. Lowest-quartile cities near internal borders show no significant decline (+0.305, $p = 0.216$), while all three upper quartiles, each with manufacturing shares above 0.75, decline by 0.5 to 0.8 pp, individually and jointly significant ($p = 0.007$). The effects do not rise from Q2 to Q4, which suggests a threshold rather than a dose-response.

At external borders Q4 cities decline more than Q1 (-0.923 vs. -0.255 pp), but the quartile interactions are imprecise (joint test $p = 0.691$). Less sectoral variation is expected there, since a truncated hinterland penalizes a city whatever it produces, whereas a severed supply chain penalizes the sectors that relied on it.

Columns (3)–(4) turn to Central Asia. The manufacturing split is uninformative there, since only 2 of the 11 treated external border cities fall in the high group ($p = 0.997$; Appendix Table A12), whereas the agriculture split gives cells of 8 high and 3 low. Low-agriculture external border cities grew 3.27 pp faster ($p = 0.006$), high-agriculture ones showed no significant gain (0.647, $p = 0.324$), and the difference is significant ($p = 0.047$). With three treated cities in the low group, this is indicative rather than precise. Internal border effects in Central Asia are insignificant across all sectors.

[9]Summary statistics for the matched sample, including the distribution of cities across heterogeneity bins by border type, are reported in Appendix Table A11.

Panel A of Table A12 reports agriculture and mining. Extraction is the clearest case of a low share. Mining cities do buy inputs, including equipment, explosives, fuel and timber, but they buy them intermittently from a small number of suppliers, whereas an assembly plant needs continuous delivery of hundreds of specific components and halts when any one fails to arrive. This is the distinction Blanchard and Kremer (1997) identify, with output falling farthest for goods with the most complex production processes. The customer side runs the same way: mining output was a standardized commodity with many interchangeable buyers, so a buyer lost across a new border could be replaced, whereas a component maker's buyers were few and specific to it (Rauch, 1999). Consistent with it, European cities with any mining employment show no decline at internal borders (+0.059, $p = 0.762$) while cities without mining decline by 0.578 pp ($p < 0.001$), and the difference is significant at $p = 0.011$. The same ordering appears at external borders (−0.170 against −0.915 pp, $p = 0.090$). The mining and manufacturing splits are related but not identical: about half of the cities with mining employment fall in the lowest manufacturing quartile, where they make up 59 percent of cities, and the other half have manufacturing shares above 0.75.

Agriculture behaves differently. European cities with agricultural employment decline by 0.307 pp against 0.446 pp for those without, a difference not distinguishable from zero ($p = 0.570$), so there is no evidence that agriculture insulates European cities as mining does. This is plausible because Soviet agro-processing was embedded in cross-republican supply chains for machinery, fertiliser and onward processing. In Central Asia the sector matters instead through the demand margin (Table 2, Column 4): what separates cities is not whether a sector is primary or industrial but which margin of market access the 1991 rupture moved for it.

Table A13 reports the two remaining sectors in the 1989 data, utilities and industrial repair; construction, retail and public administration are empty or near-empty. Utilities, mostly electric power, gas transmission, district heating and water supply, sold much of their output locally but relied on fuel and network links that crossed republican boundaries, so the split does not isolate either margin of market access. European cities with utility employment show no detectable border decline at either frontier (−0.131 and −0.018 pp), against declines of 0.585 and 1.088 pp for cities without them ($p = 0.090$ and $p = 0.035$), although utility-employing cities grew 0.556 pp more slowly after 1991 regardless of border proximity. Cities with and without repair employment decline similarly (−0.413 against −0.368 pp, $p = 0.854$). I report both splits for completeness rather than as tests of the mechanism.

Alternative Accounts

I address compositional alternatives by selecting comparison samples through propensity-score matching. Matching is conducted separately by region and border type on the city cross-section. Probit specifications use 1970 population, 1989 industrial employment, turnover, or manufacturing output separately, or defence enterprise count and workforce jointly. Each treated city is matched, with replacement and under common support, to up to two cities unexposed to that border type, chosen by proximity in propensity score. I then re-estimate the baseline difference-in-differences on the selected cities' full population panels. Each selected

city enters once per period, without weighting controls by their frequency of selection. An exact-match restriction retains cities in 1989 SIC categories containing both treated and control observations. Additional European specifications match jointly on all six covariates, with variants restricting eligible controls to within 500 or 1,000 km of each treated city.

Table 3 reports the European estimates for the six matching exercises described above.[10] Internal border declines range from 0.36 to 0.58 pp, bracketing the baseline estimate of 0.40 pp, and external border declines from 0.53 to 0.90 pp, all statistically significant, indicating that the estimates are robust to these matching restrictions.[11]

Column (6) addresses military divestment specifically. The matched sample is smaller (267 cities for internal borders, 100 for external) but the border effects persist at similar magnitudes: −0.46 pp for internal borders and −0.83 pp for external borders, both statistically significant. The Central Asian matched samples (Appendix Table A15) contain only 5 to 25 cities and support no systematic inference.

Table A16 varies the matching design itself. Column (1) conditions jointly on all six covariates, yielding −0.505 pp for internal borders on the 1,076 cities with complete covariate data, and Column (4) gives −0.802 pp for external borders. Balance in these matched samples is close: mean absolute standardized bias falls from 11.5 to 4.1 percent, and a probit of treatment on all six covariates in the matched sample has a pseudo R^2 of 0.001 and cannot reject that the two groups are identical ($\chi^2(6) = 1.17$, $p = 0.978$).

Figure A4 maps the matches. Selected controls are geographically dispersed, but proximity in propensity score does not ensure geographic proximity: the median pair distance is 1,326 km. Following Redding and Sturm (2008), who impose a geographic band on their control group, Columns (2)–(3) and (5)–(6) of Table A16 require each control to lie within 500 or 1,000 km of its treated city. Every treated city retains two matched controls even at 500 km, where the median pair distance falls to 347 km, and the estimates are stable: −0.392 and −0.600 pp for internal borders and −0.526 and −0.699 pp for external borders, against baseline estimates of −0.402 and −0.595 pp.

Finally, Soviet Science Cities and other closed settlements lay overwhelmingly in the interior, and Schweiger et al. (2022) show that Science Cities outperformed comparable localities after 1991, so as controls they could inflate the estimated border penalties. Excluding sample cities within 3 to 20 km of the 91 locatable Science Cities, and 20 further closed or restricted-access cities elsewhere in the union, leaves the European internal effect between −0.361 and −0.395 pp and the external effect between −0.592 and −0.609 pp, and the Central Asian external gain at 0.833 pp, significant at the 10% level (Table A17).

[10] Appendix Table A14 reports covariate balance.

[11] Synthetic control, the natural alternative in border settings, is built for a few treated units observed over a long pre-treatment series, whereas there are 353 treated cities along internal frontiers in post-Soviet Europe and three pre-1991 census observations, and the object of interest is an average effect across a large treated group.

6 Conclusion

This paper examines how geopolitical fragmentation affects urban growth, exploiting the Soviet collapse as a natural experiment across nearly 1,400 cities and a five-decade horizon. In post-Soviet Europe, cities near the new internal borders and the old external frontier experienced persistent population declines relative to interior cities; in Central Asia, cities near previously sealed external borders grew faster. The distance, sectoral, size, and coastal patterns are consistent with market access as an important channel.

The spatial consequences of state dissolution extend beyond newly created boundaries: changes in allocation institutions can alter the economic importance of existing frontiers, exposing disadvantages previously buffered by administrative coordination. Border changes are not mechanically harmful or beneficial for nearby places: where a boundary cuts through integrated production networks, nearby cities experience persistent relative decline, and where it unlocks access to previously closed markets they can experience sustained relative growth. Because the effects concentrate among smaller cities, geopolitical fragmentation widens spatial inequality, disproportionately penalizing cities least equipped to absorb the loss of cross-border market access. As movements for regional autonomy continue globally, this asymmetry deserves attention from policymakers weighing the spatial economic costs of fragmentation.

Figures & Tables



Figure 1: City Types and Regional Classification in the Former Soviet Union

Notes: The map shows all 1,396 cities in the sample. Cities are classified into four mutually exclusive categories by border exposure: within 75 km of a former inter-republican (internal) border only, within 75 km of an external border only, within 75 km of both, and interior cities beyond 75 km of any border. Blue lines mark internal borders, the boundaries between post-Soviet states that were administrative before 1991; black lines mark external borders, the frontiers that were already international throughout the Soviet period. Fills distinguish the two regional groupings used in the analysis: post-Soviet Europe comprises Armenia, Azerbaijan, Belarus, Estonia, Georgia, Latvia, Lithuania, Moldova, Russia, and Ukraine; Central Asia comprises Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan, and Uzbekistan, following Council of the European Union (2009). Russia is classified with post-Soviet Europe despite spanning both continents: the grouping reflects the density of cross-republican production linkages rather than physical geography, and Russian cities in Asia are treated on the same basis as all others, facing internal frontiers with Kazakhstan and external frontiers with China and Mongolia. A small number of far-eastern Russian cities in the estimation sample lie outside the plotted extent. The 75 km cut-off follows Redding and Sturm (2008).

Alt text: Map of the former Soviet Union showing the 1,396 sample cities, with markers distinguishing interior cities and cities near an internal border only, an external border only, or both. Internal borders between post-Soviet states are drawn in blue and external frontiers in black; post-Soviet Europe, including Russia, and Central Asia are shaded in different colors. Most border cities lie in the western republics, the Caucasus, and Central Asia, while most interior cities are in Russia.

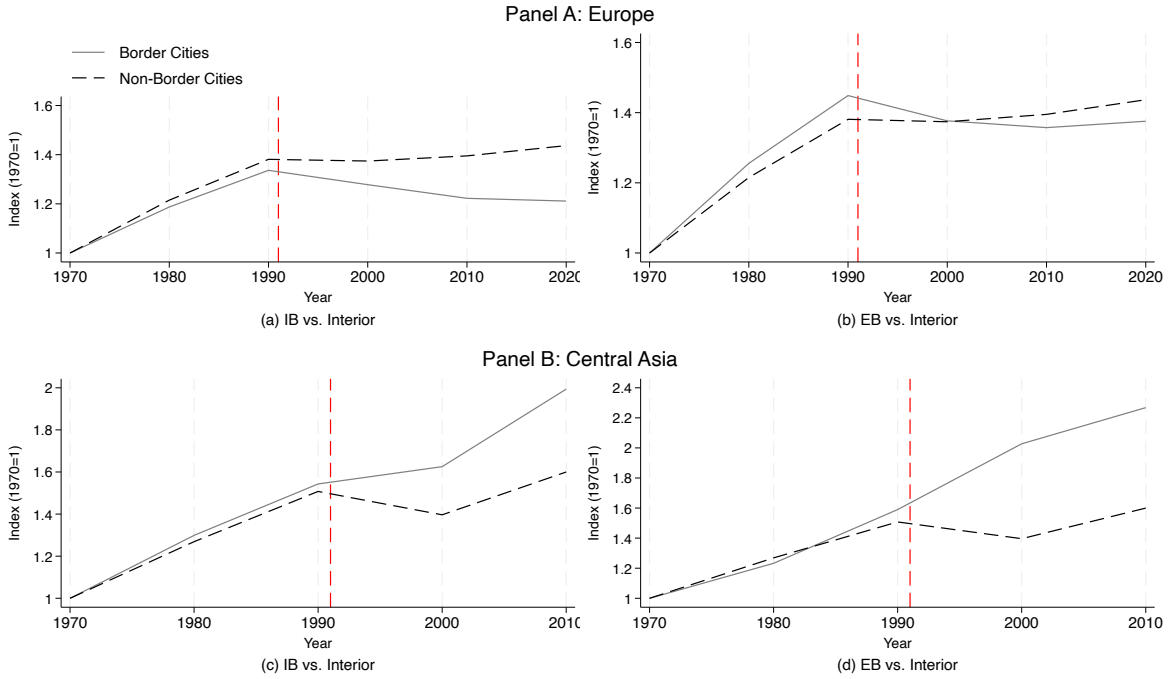


Figure 2: Indices of Border and Non-Border City Population

Notes: Panel A includes all post-Soviet countries geographically situated in Greater Europe: Armenia, Azerbaijan, Belarus, Estonia, Georgia, Latvia, Lithuania, Russia, and Ukraine. Moldova is excluded due to an unbalanced panel. The Central Asia panel includes all five post-Soviet Central Asian countries, but the balanced index shown here includes only two post-collapse periods, as Turkmenistan and Uzbekistan conducted only two censuses after the Soviet dissolution.

Alt text: Four line charts of total population, indexed to 1 in 1970, for border and interior cities. In post-Soviet Europe, border and interior cities grow in parallel until 1991; afterwards interior cities keep growing while cities near internal and external borders stagnate or decline. In Central Asia, border cities grow faster than interior cities after 1991, most visibly near external borders.

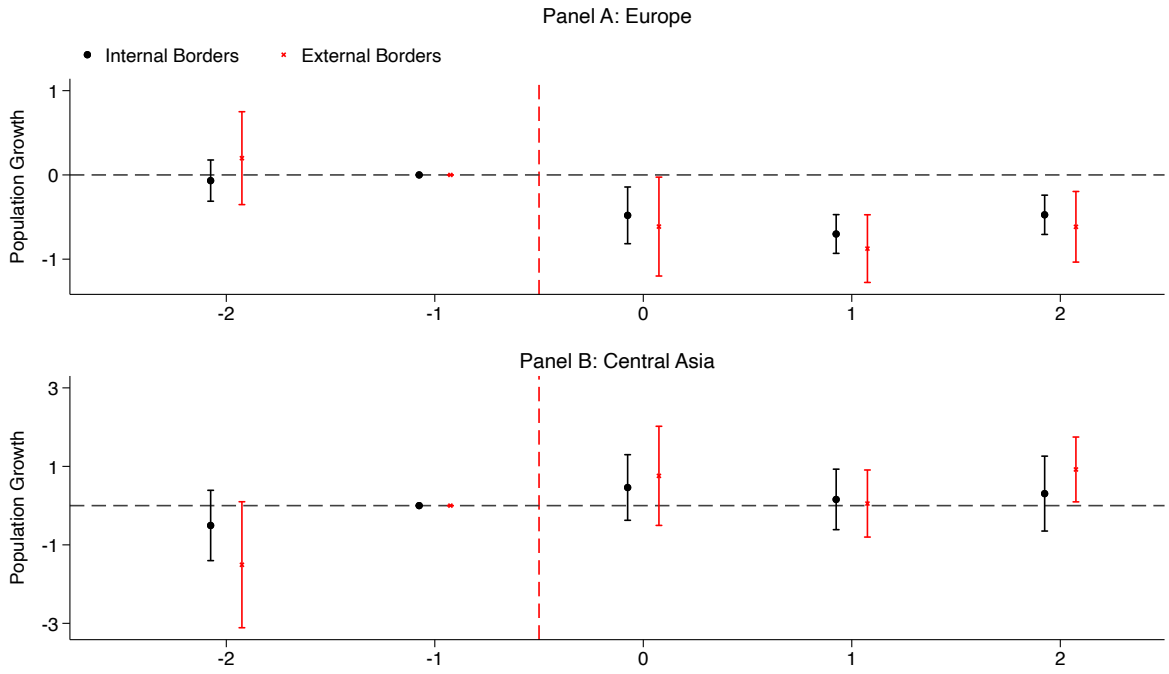


Figure 3: Event-Study Estimates: Baseline

Notes: Coefficient plots from event-study regressions of annualized population growth on border proximity interacted with census-period indicators (equation 6). Black circles denote internal borders; red crosses denote external borders. Panel A shows Europe; Panel B shows Central Asia. The omitted period is $k = -1$ (1979–1989). All specifications include city and period fixed effects. Control group: interior cities beyond 75 km from any border. Standard errors clustered at the city level; bars show 95% confidence intervals.

Alt text: Two event-study plots of coefficients with 95% confidence intervals for cities near internal and external borders, by census period relative to the 1991 dissolution. In post-Soviet Europe, coefficients are close to zero before 1991 and negative afterwards for both border types. In Central Asia, post-1991 coefficients are mostly positive but imprecise, and the external-border coefficient two periods before dissolution is negative.

Table 1: Urban Growth and Market Access: Main Results

	Dependent Variable: Population Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
IB × Division	−0.210**	−0.397***	−0.402***	0.471		
	(0.105)	(0.111)	(0.111)	(0.370)		
EB × Division	−0.563***	−0.591***	−0.595***	0.969**		
	(0.190)	(0.200)	(0.200)	(0.451)		
CA × Division		0.079				
		(0.323)				
IB × Division × CA		0.808**				
		(0.378)				
EB × Division × CA		1.373***				
		(0.480)				
IB only × Division					−0.358***	0.552
					(0.113)	(0.395)
EB only × Division					−0.441	1.356**
					(0.291)	(0.620)
Both IB & EB × Division					−1.099***	0.803
					(0.266)	(0.520)
Implied IB effect, CA		0.411				
p-value		0.256				
Implied EB effect, CA		0.781				
p-value		0.074				
p: CA interactions jointly zero		0.006				
Both − IB only					−0.740	0.251
p-value					0.007	0.564
Both − EB only					−0.657	−0.553
p-value					0.088	0.388
p: effects additive					0.456	0.143
Outcome mean (%)	0.676	0.676	0.583	1.369	0.583	1.369
Sample	All	All	Europe	CA	Europe	CA
Cities: IB only	373	373	273	100	273	100
Cities: EB only	72	72	62	10	62	10
Cities: both	85	85	80	5	80	5
Cities: interior	866	866	806	60	806	60
Cities	1,396	1,396	1,221	175	1,221	175
Observations	6,809	6,809	6,004	805	6,004	805
City FE	✓	✓	✓	✓	✓	✓
Period FE	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is annualized city population growth (%). "IB" and "EB" denote cities less than 75 km from internal and external borders, respectively; "CA" denotes Central Asia (Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan, and Uzbekistan). "Division" equals one for post-1991 periods. All columns include city and period fixed effects. Columns (1)–(2) pool all 1,396 cities. Column (1) imposes common border coefficients across regions. Column (2) adds the Central Asia indicator interacted with Division and with each border treatment: the IB and EB coefficients refer to post-Soviet Europe, and the triple interactions report Central Asian differences relative to Europe. Implied Central Asian effects are the corresponding coefficient sums, with *p*-values from Wald tests; the joint test evaluates whether both triple interactions equal zero. Columns (3)–(4) estimate the baseline separately for Europe and Central Asia. In Columns (1)–(4), both border indicators equal one for dual-exposure cities. Columns (5)–(6) instead distinguish mutually exclusive internal-only, external-only, and dual-exposure categories, with interior cities as the omitted group. "Both − IB only" and "Both − EB only" compare dual exposure with each single-exposure category. The additivity test evaluates whether the dual-exposure coefficient equals the sum of the two single-exposure coefficients. City counts report unique cities in each column's estimation sample; the four exposure categories sum to the total number of cities. Standard errors clustered at the city level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Urban Growth and Sectoral Composition

	Dependent Variable: Population Growth			
	Europe: Manufacturing Quartiles		Central Asia: Agriculture Median Split	
	(1) Internal	(2) External	(3) Internal	(4) External
Border × Division (Q1, ref.)	0.305	−0.255		
	(0.247)	(0.520)		
Border × Division × Q2	−1.102***	−0.313		
	(0.335)	(0.627)		
Border × Division × Q3	−0.912***	−0.659		
	(0.334)	(0.619)		
Border × Div. × Q4 (highest)	−0.813**	−0.668		
	(0.368)	(0.738)		
Q2 × Division	0.460**	0.460**		
	(0.184)	(0.184)		
Q3 × Division	0.491***	0.491***		
	(0.183)	(0.183)		
Q4 × Division	0.474**	0.474**		
	(0.228)	(0.228)		
Border × Div. × High Agric.			0.554	0.647
			(0.584)	(0.651)
Border × Div.× Low Agric.			0.340	3.268***
			(0.642)	(1.135)
High Agric. × Division			0.273	0.259
			(0.782)	(0.789)
p: joint Q2=Q3=Q4=0	0.007	0.691		
p: Q4 vs Q1	0.028	0.366		
p: High = Low			0.805	0.047
Outcome mean (%)	0.527	0.578	1.362	1.328
Cities	1,021	848	141	61
Observations	5,033	4,206	652	284
City FE	✓	✓	✓	✓
Period FE	✓	✓	✓	✓

Notes: The dependent variable is annualized city population growth (%). The matched industry sample contains 1,076 European cities. Columns (1)–(2) exclude cities near external borders only (*ext_only*=1), so that the internal border effect is identified against interior cities not exposed to external border shocks. Columns (3)–(4) similarly exclude cities near internal borders only. This accounts for the difference between the 1,076 matched cities and the column-specific sample sizes. Columns (1)–(2) decompose the European border effect by 1989 manufacturing employment quartiles, where Q1 (lowest manufacturing share) is the reference category. The quartile cutpoints are computed over all European cities with sectoral data. Column (1) estimates the internal border effect; Column (2) estimates the external border effect. Columns (3)–(4) split Central Asian border cities at the within-region median of 1989 agricultural employment share; "High" denotes cities above the median and "Low" denotes cities at or below the median. Because the median agricultural share is close to zero, the split effectively compares cities with any substantial agricultural employment to those without. "Border" refers to cities within 75 km of an internal border in Columns (1) and (3) and an external border in Columns (2) and (4). "Division" equals one for post-1991 periods. Columns (1) and (2) exclude cities near external and internal borders only, respectively. Standard errors clustered at the city level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Robustness to Matching on Pre-1991 Characteristics (Europe)

	Dependent Variable: Population Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Internal Borders						
IB × Division	−0.576***	−0.361**	−0.571***	−0.443***	−0.381***	−0.463**
	(0.128)	(0.143)	(0.129)	(0.136)	(0.125)	(0.199)
Matching on	1970 Pop.	1989 Ind. Empl.	1989 Ind. Turn.	1989 Manuf.	1989 SIC	Military
Outcome mean (%)	0.568	0.542	0.457	0.471	0.512	0.537
Cities	737	655	651	657	945	267
Observations	3,601	3,207	3,187	3,217	4,654	1,309
Adj. R^2	0.427	0.432	0.433	0.436	0.454	0.474
Panel B: External Borders						
EB × Division	−0.528**	−0.690***	−0.901***	−0.623**	−0.626***	−0.826**
	(0.247)	(0.258)	(0.248)	(0.252)	(0.237)	(0.409)
Matching on	1970 Pop.	1989 Ind. Empl.	1989 Ind. Turn.	1989 Manuf.	1989 SIC	Military
Outcome mean (%)	0.747	0.637	0.574	0.564	0.566	0.745
Cities	310	254	278	275	752	100
Observations	1,512	1,243	1,360	1,346	3,726	490
Adj. R^2	0.389	0.365	0.379	0.397	0.439	0.309
City FE	✓	✓	✓	✓	✓	✓
Period FE	✓	✓	✓	✓	✓	✓

Notes: Each column re-estimates the baseline specification on a matched subsample of treated border cities and their nearest neighbors. "Cities" in the matching tables reports the number of unique cities in the matched sample; "Observations" equals cities multiplied by census periods. Columns (1)–(4) and (6) use 1:2 nearest-neighbor propensity score matching with replacement on the city cross-section: each treated (border) city is paired with its two closest interior cities based on probit-estimated propensity scores for the indicated covariate. Because matching is with replacement, a control city may serve as a neighbor for multiple treated cities but enters the estimation sample once. Column (5) uses exact matching on each city's primary SIC code, restricting the sample to border and interior cities that shared the same dominant industrial classification in 1989; the larger sample reflects the retention of all cities within shared SIC strata rather than a fixed number of neighbors per treated city. "Military" matches jointly on the count and total workforce of defense enterprises that closed between 1989 and 1991. The dependent variable is annualized city population growth (%). "Division" equals one for post-1991 periods. All specifications include city and period fixed effects. Standard errors clustered at the city level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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Appendix A - Figures & Tables

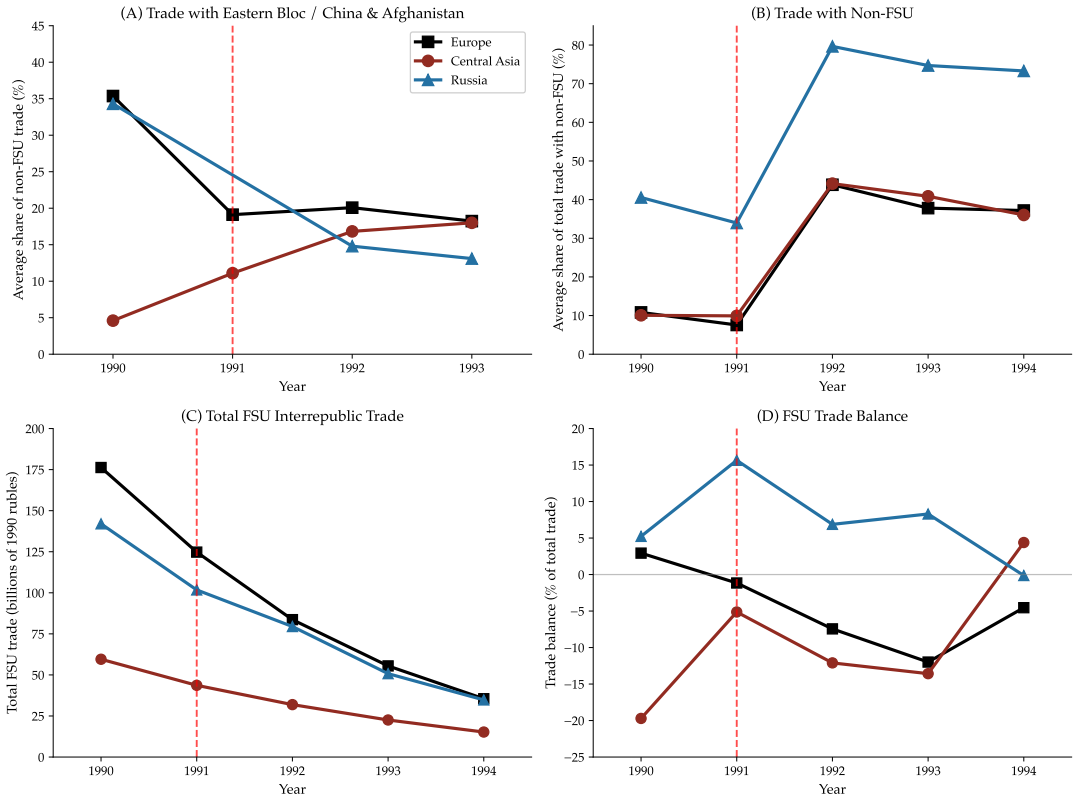


Figure A1: Post-Soviet Trade Realignment: Europe vs. Central Asia

Notes: This figure presents four panels illustrating trade reorientation patterns among Former Soviet Union (FSU) republics from 1990–1994. Countries are grouped into three regions: "Europe" (Armenia, Azerbaijan, Belarus, Estonia, Georgia, Latvia, Lithuania, Moldova, Ukraine; $n = 9$), "Central Asia" (Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan, Uzbekistan; $n = 5$), and Russia (shown separately). The vertical dashed line in each panel marks December 1991, when the USSR formally dissolved. The partner groups are defined by adjacency: the Eastern Bloc lay across the western frontier of the European republics, and China and Afghanistan across Central Asia's southern and eastern frontiers, so in each region the group shown is the set of non-Soviet economies whose frontier with that region changed status in 1991. Both groups cannot be shown for both regions on a consistent basis, since Belkindas and Ivanova (1995) report partner-country trade unevenly across the Central Asian republics. **Panel A** shows the average share of non-FSU trade conducted with Bulgaria, Czechoslovakia, Hungary, Poland, and Romania for European republics and Russia, and China plus Afghanistan for Central Asian republics. Turkmenistan is dropped due to lack of data. Regional values represent unweighted country averages. **Panel B** shows the average share of each region's total trade (exports plus imports) conducted with non-FSU countries (i.e., rest of world). This measure captures the overall reorientation away from intra-FSU trade toward global markets. Data are from IMF Direction of Trade Statistics and national sources. Regional values represent unweighted country averages. **Panel C** shows total interrepublic trade within the FSU, measured in billions of constant 1990 rubles to enable comparison across years despite hyperinflation that ensued. For each region, total trade equals the sum of exports and imports with all other FSU republics. **Panel D** shows the trade balance as a percentage of total FSU trade. Positive values indicate a trade surplus with other FSU republics; negative values indicate a deficit. Raw data source is Belkindas and Ivanova (1995). Coverage ends in 1994 because the Soviet-format interrepublic trade accounts on which Panels C and D rely were discontinued once the successor states began reporting as independent customs territories; no consistent constant-ruble series is available thereafter. The figure documents the sharpness and regional asymmetry of the initial reorientation rather than the trajectory of trade over the full study period.

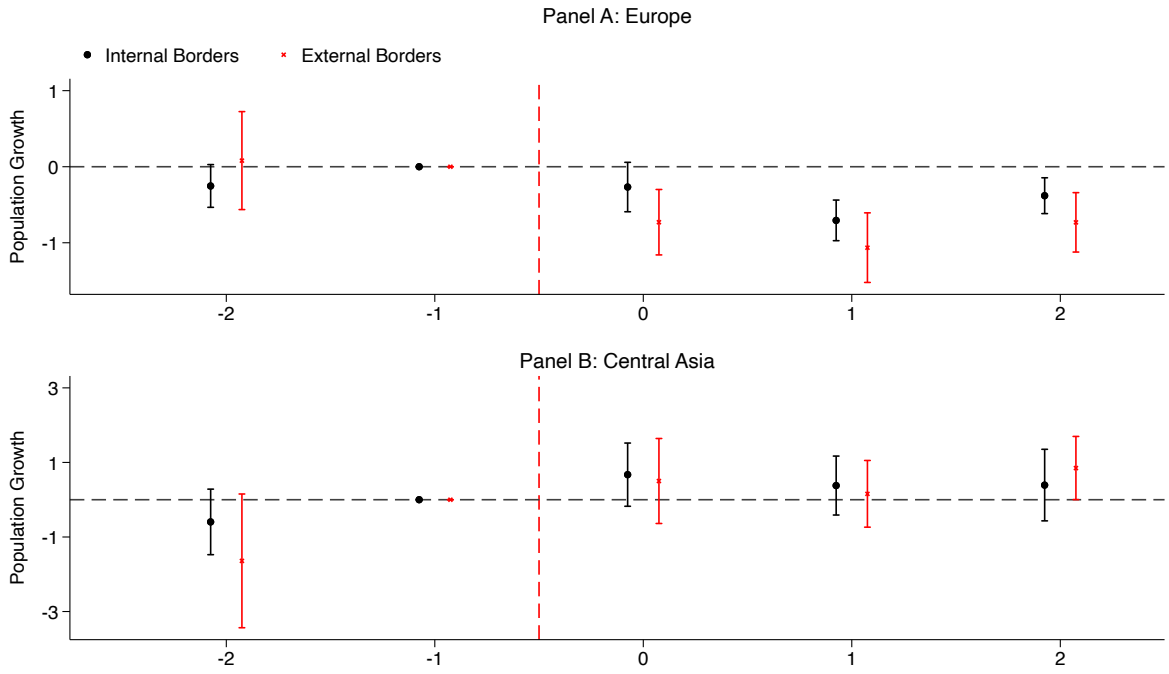
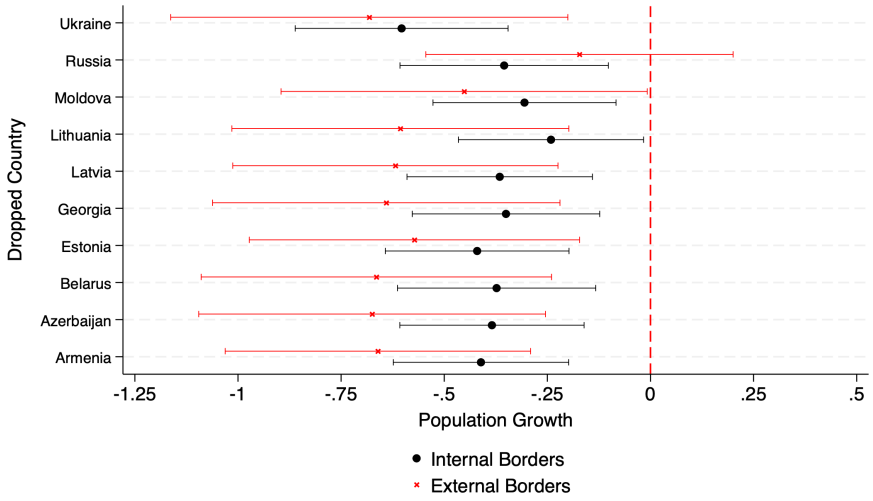


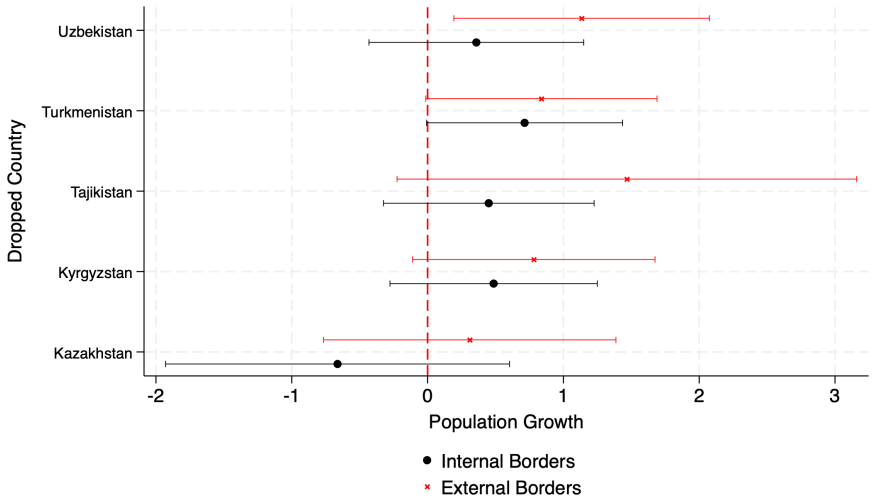
Figure A2: Event-Study Estimates: Sub-Region-by-Period Fixed Effects

Notes: This figure replicates Figure 3 with more demanding fixed effects. Panel A replaces period fixed effects with sub-region-by-period fixed effects, where sub-regions are Baltics, Caucasus, and Eastern Europe. Panel B excludes Turkmenistan, whose census timing (1995, 1999) differs substantially from the other Central Asian countries. All other specifications are identical to Figure 3. Standard errors clustered at the city level; bars show 95% confidence intervals.



(a) Europe

Notes: Each point represents the estimated coefficient from a specification that drops the labeled country. Circles denote IB effects; X markers denote EB effects. Both IB and EB coefficients remain negative across all exclusions. Precision decreases when dropping Russia, which contributes the largest share of cities to the European sample. Bars show 95% confidence intervals.



(b) Central Asia

Notes: Each point represents the estimated coefficient from a specification that drops the labeled country. The EB coefficient remains positive in all five exclusions, though precision varies. The estimate is most attenuated when dropping Kazakhstan, which provides the largest share of interior control cities. The IB coefficient turns negative when Kazakhstan is excluded, again due to the loss in the share of interior control cities. Bars show 95% confidence intervals.

Figure A3: Leave-One-Out Analysis

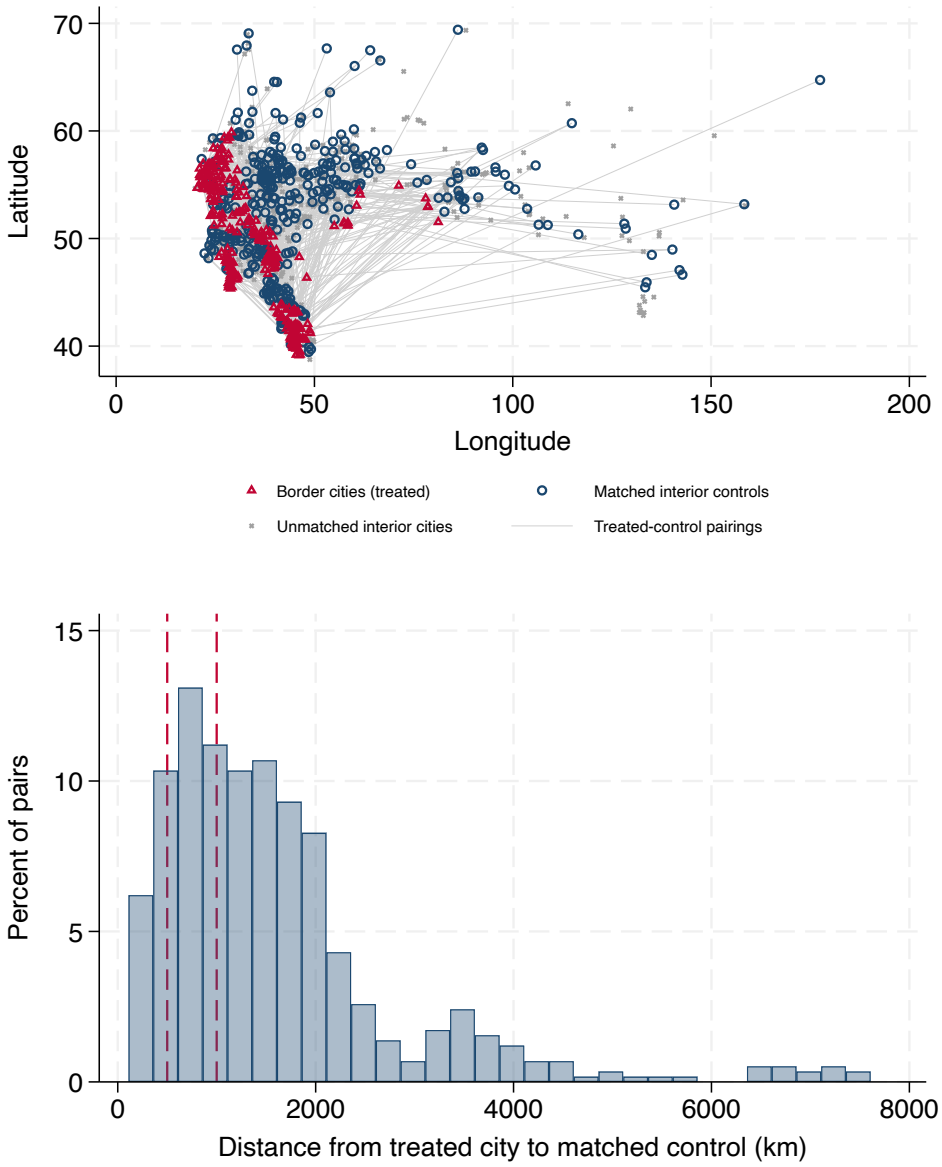


Figure A4: Matching Diagnostics: Internal Borders, Post-Soviet Europe

Notes: Diagnostics for the multivariate propensity score match reported in Column (1) of Table A16, which conditions on 1970 population, 1989 industrial employment, turnover, and manufacturing output, and the count and workforce of defence enterprises closing between 1989 and 1991. Matching is 1:2 nearest-neighbour with replacement under common support, with the score estimated by probit on the city cross-section. The sample is the 1,076 post-Soviet European cities with complete covariate data, fewer than the 1,221 in the baseline because the 1989 industrial data do not cover every city. **Panel A** plots each of the 290 treated border cities on support, the interior cities selected as its two matched controls, and the pairings between them; interior cities not selected as controls are shown separately. **Panel B** plots the distribution of great-circle distance between each treated city and its matched controls, with the 500 and 1,000 km radii imposed in Table A16, Columns (2)–(3) marked. The median unconstrained pair distance is 1,326 km, since the propensity score is indifferent to location; every treated city nonetheless retains two matched controls when a 500 km radius is imposed. Covariate balance is reported in Table A14.

Table A1: Summary of Census Data

Country	Cities	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Armenia	29	1970	1979	1989	2001	2011	2016
Azerbaijan	43	1970	1979	1989	2002	2011	2015
Belarus	80	1970	1979	1989	1999	2009	2019
Estonia	21	1970	1979	1989	2000	2011	2017
Georgia	41	1970	1979	1989	2002	2014	2020
Kazakhstan	77	1970	1979	1989	1999	2009	2021
Kyrgyzstan	31	1970	1979	1989	1999	2009	2021
Latvia	23	1970	1979	1989	2000	2011	2020
Lithuania	41	1970	1979	1989	2001	2011	2021
Moldova	37	1970	1979	1989	2004	2014	
Russia	702	1970	1979	1989	2002	2010	2021
Tajikistan	29	1970	1979	1989	2000	2010	2021
Turkmenistan	9	1970	1979	1989	1995	1999	
Ukraine	204	1970	1979	1989	2001	2014	2021
Uzbekistan	29	1970	1979	1989	2005	2020	
Total	1396						

Notes: Census years are recorded for each country from 1970 to the most recent available year. Years 1–3 correspond to Soviet-wide censuses (1970, 1979, 1989), which are common across all countries. Years 4–6 correspond to post-independence national censuses, whose timing varies by country.

Table A2: Treatment Variable Statistics by Region

	Mean	[SD]	Min	Max	Treated	Control	Obs.
Panel A: Europe							
Outcome variable:							
Annualized Pop. Growth	0.583	[1.940]	-13.61	27.25	1221 cities		6,004
Main treatment variables:							
Post-1991 (Division)	0.593	[0.491]	0	1	1221	1221	6,004
Internal border (<75 km)	0.281	[0.449]	0	1	353	868	6,004
External border (<75 km)	0.112	[0.316]	0	1	142	1079	6,004
IB × Division	0.163	[0.369]	0	1	353	1221	6,004
EB × Division	0.065	[0.246]	0	1	142	1221	6,004
Distance bins (mutually exclusive):							
IB 0–25 km	0.084	[0.277]	0	1	107	775	6,004
IB 25–50 km	0.111	[0.314]	0	1	139	775	6,004
IB 50–75 km	0.086	[0.280]	0	1	107	775	6,004
IB 75–100 km	0.076	[0.264]	0	1	93	775	6,004
EB 0–25 km	0.037	[0.188]	0	1	47	1036	6,004
EB 25–50 km	0.037	[0.190]	0	1	47	1036	6,004
EB 50–75 km	0.038	[0.191]	0	1	48	1036	6,004
EB 75–100 km	0.033	[0.179]	0	1	43	1036	6,004
Panel B: Central Asia							
Outcome variable:							
Annualized Pop. Growth	1.369	[2.301]	-8.71	16.36	175 cities		805
Main treatment variables:							
Post-1991 (Division)	0.565	[0.496]	0	1	175	175	805
Internal border (<75 km)	0.591	[0.492]	0	1	105	70	805
External border (<75 km)	0.076	[0.265]	0	1	15	160	805
IB × Division	0.330	[0.471]	0	1	105	175	805
EB × Division	0.039	[0.193]	0	1	15	175	805
Distance bins (mutually exclusive):							
IB 0–25 km	0.323	[0.468]	0	1	58	61	805
IB 25–50 km	0.171	[0.377]	0	1	30	61	805
IB 50–75 km	0.097	[0.296]	0	1	17	61	805
IB 75–100 km	0.050	[0.217]	0	1	9	61	805
EB 0–25 km	0.045	[0.207]	0	1	9	156	805
EB 25–50 km	0.011	[0.105]	0	1	2	156	805
EB 50–75 km	0.020	[0.140]	0	1	4	156	805
EB 75–100 km	0.022	[0.148]	0	1	4	156	805

Notes: All statistics are computed on the estimation panel (city × census-period observations with non-missing annualized population growth), which matches the main regression samples. All treatment variables are binary indicators. "Treated" and "Control" report the number of unique cities for which the indicator equals one or zero. For time-invariant indicators (e.g., Internal border), the two columns sum to the total number of cities in the region. For interaction terms (e.g., IB × Division), "Treated" counts border cities, which have the indicator equal to one only in post-1991 periods, while "Control" counts all cities that ever have the indicator equal to zero, including border cities in their pre-1991 periods; the two columns therefore overlap and do not sum to the total. For distance bins, "Control" reports interior cities (>100 km from the relevant border type), the omitted reference category in the gradient regressions (Table A9). Distance bins are mutually exclusive 25 km bands. The outcome variable row reports total unique cities in the estimation sample rather than a treated/control split.

Table A3: Descriptive Statistics by Region

Variable	Group	Mean	SD	Min	Max	Obs.
			Europe			
Population	IB=1	76,435	167,631	3000	1,609,959	2,084
	IB=0	116,223	417,391	3059	12,455,682	5,205
1970 Population	IB=1	61,305	136,566	5067	1,222,852	2,118
	IB=0	88,995	299,802	5084	6,941,961	5,208
IB Distance (km)	IB=1	38.48	20.66	0.080	74.52	2,118
	IB=0	460.7	664.9	75.50	5078	5,208
EB Distance (km)	IB=1	311.9	314.0	0.516	1659	2,118
	IB=0	694.5	499.6	0.066	3034	5,208
Large (=1)	IB=1	0.368	0.482	0.000	1.000	2,118
	IB=0	0.566	0.496	0.000	1.000	5,208
Population	EB=1	66,605	140,177	3000	1,201,539	826
	EB=0	109,734	383,409	3059	12,455,682	6,463
1970 Population	EB=1	49,120	103,983	5067	766,705	852
	EB=0	85,184	277,502	5084	6,941,961	6,474
IB Distance (km)	EB=1	341.9	823.6	0.080	3334	852
	EB=0	338.2	555.1	0.259	5078	6,474
EB Distance (km)	EB=1	37.47	22.33	0.066	74.85	852
	EB=0	655.8	471.7	75.36	3034	6,474
Large (=1)	EB=1	0.282	0.450	0.000	1.000	852
	EB=0	0.538	0.499	0.000	1.000	6,474
			Central Asia			
Population	IB=1	108,327	252,647	4351	2,571,668	581
	IB=0	81,422	114,737	6010	1,184,469	399
1970 Population	IB=1	70,553	162,565	5053	1,384,509	630
	IB=0	55,322	76,512	6010	523,271	420
IB Distance (km)	IB=1	25.86	19.86	0.655	70.68	630
	IB=0	204.5	108.2	78.52	502	420
EB Distance (km)	IB=1	298.0	280.8	6.290	1540	630
	IB=0	560.8	409.1	2.824	1464	420
Large (=1)	IB=1	0.390	0.488	0.000	1.000	630
	IB=0	0.514	0.500	0.000	1.000	420
Population	EB=1	55,059	107,879	5053	604,700	76
	EB=0	100,930	214,151	4351	2,571,668	904
1970 Population	EB=1	32,674	60,280	5053	253,118	90
	EB=0	67,441	139,683	5239	1,384,509	960
IB Distance (km)	EB=1	134.1	100.0	26.04	382	90
	EB=0	93.88	112.7	0.655	502	960
EB Distance (km)	EB=1	26.35	21.04	2.824	59.52	90
	EB=0	438.5	358.3	80.53	1540	960
Large (=1)	EB=1	0.267	0.445	0.000	1.000	90
	EB=0	0.456	0.498	0.000	1.000	960
			Europe			
Log(Lights)	IB=1	7.524	1.735	0.693	11.95	6,072
	IB=0	8.640	1.480	0.693	14.50	13,332
Log(Lights)	EB=1	7.177	1.588	0.693	11.13	2,662
	EB=0	8.468	1.586	0.693	14.50	16,742
			Central Asia			
Log(Lights)	IB=1	8.105	1.751	0.693	12.05	1,496
	IB=0	8.134	1.616	3.178	11.54	1,166
Log(Lights)	EB=1	6.945	1.720	3.178	11.50	308
	EB=0	8.271	1.628	0.693	12.05	2,354

Notes: Statistics are reported on the full city × census-period panel, which includes six periods per city (three pre-1991, three post-1991). Population is the level recorded at each census. 1970 Population is time-invariant but reported for all city-periods. IB and EB Distance measure straight-line kilometers to the nearest internal and external post-Soviet border. Large is a binary indicator for cities above the 1970 sample median population. Log(Lights) is from a separate annual city-year panel (1992–2013) matched to the same city universe; observation counts reflect city × year. Treatment variable statistics and the outcome variable (annualized population growth) are reported in Table A2, computed on the estimation sample (city-period observations with non-missing growth rates).

Table A4: Robustness to Sub-Region-by-Period Fixed Effects and Sample Restrictions

	Dependent Variable: Population Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
IB × Division	−0.159	−0.241**	−0.241**	0.714*		
	(0.112)	(0.114)	(0.114)	(0.368)		
EB × Division	−0.540***	−0.663***	−0.663***	0.838*		
	(0.184)	(0.196)	(0.196)	(0.434)		
IB × Division × CA		0.712*				
		(0.386)				
EB × Division × CA		1.633***				
		(0.490)				
IB only × Division					−0.172	0.778**
					(0.121)	(0.393)
EB only × Division					−0.437	1.192*
					(0.305)	(0.629)
Both IB & EB × Division					−1.047***	1.078**
					(0.244)	(0.518)
Implied IB effect, CA		0.471				
p-value		0.201				
Implied EB effect, CA		0.969				
p-value		0.031				
p: CA interactions jointly zero		0.002				
Both − IB only					−0.875	0.301
p-value					0.001	0.492
Both − EB only					−0.610	−0.113
p-value					0.105	0.862
p: effects additive					0.279	0.242
Outcome mean (%)	0.676	0.676	0.583	1.309	0.583	1.309
Sample	All	All	Europe	CA (exc. Turk.)	Europe	CA (exc. Turk.)
Cities: IB only	373	373	273	98	273	98
Cities: EB only	72	72	62	8	62	8
Cities: both	85	85	80	5	80	5
Cities: interior	866	866	806	55	806	55
Cities	1,396	1,396	1,221	166	1,221	166
Observations	6,809	6,809	6,004	769	6,004	769
City FE	✓	✓	✓	✓	✓	✓
Sub-Region-by-Period FE	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is annualized city population growth (%). "IB" and "EB" denote cities less than 75 km from internal and external borders, respectively; "CA" denotes Central Asia (Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan, and Uzbekistan). "Division" equals one for post-1991 periods. All columns include city and sub-region-by-period fixed effects. Sub-regions are the Baltics, the Caucasus, Slavic Europe plus Moldova, and Central Asia; within Central Asia, these time effects are equivalent to period fixed effects. Columns (1)–(2) retain the full sample of 1,396 cities, including Turkmenistan. Column (1) imposes common border coefficients across regions. Column (2) adds Central Asian interactions: the IB and EB coefficients refer to post-Soviet Europe, and the triple interactions report Central Asian differences relative to Europe. The CA-by-Division term is absorbed by the fixed effects. Implied Central Asian effects are coefficient sums, with *p*-values from Wald tests; the joint test evaluates whether both triple interactions equal zero. Columns (3)–(4) estimate the additive specification separately by region. Columns (5)–(6) distinguish mutually exclusive internal-only, external-only, and dual-exposure categories, with interior cities as the omitted group. "Both − IB only" and "Both − EB only" compare dual exposure with each single-exposure category. The additivity test evaluates whether the dual-exposure coefficient equals the sum of the two single-exposure coefficients. Columns (4) and (6) exclude Turkmenistan, whose census timing differs from the other Central Asian countries. City counts report unique cities in each column's estimation sample. Standard errors clustered at the city level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Border Effects by City Size

	Dependent Variable: Population Growth					
	All		Europe		Central Asia	
	(1)	(2)	(3)	(4)	(5)	(6)
IB × Division	−0.197	−0.024	−0.447**	−0.130	0.923	0.180
	(0.170)	(0.105)	(0.177)	(0.115)	(0.611)	(0.405)
EB × Division	−0.567**	−0.187	−0.587**	−0.200	1.402**	0.601
	(0.257)	(0.203)	(0.273)	(0.205)	(0.629)	(0.803)
City size	Small	Large	Small	Large	Small	Large
Outcome mean (%)	0.838	0.517	0.752	0.423	1.394	1.337
Treated IB cities	287	171	223	130	64	41
Treated EB cities	113	44	102	40	11	4
Cities	698	698	600	621	98	77
Observations	3,372	3,437	2,919	3,085	453	352
City FE	✓	✓	✓	✓	✓	✓
Period FE	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is annualized city population growth (%). "IB" and "EB" denote cities within 75 km of internal and external borders, respectively; "Division" equals one for post-1991 periods. Cities are split at the median 1970 population across the full sample (approximately 28,800 residents), so the cut point is common to both regions rather than computed within region. This absolute threshold is the relevant one for assessing whether the effects are concentrated in the large cities that absorbed the bulk of post-1991 return migration. The pooled cut divides each region close to evenly, placing 49 percent of European and 56 percent of Central Asian cities in the small group. Because the split partitions treated and control cities together, each column compares border to interior cities within the same size range. Odd-numbered columns report small cities and even-numbered columns large cities, for the pooled sample (Columns 1–2), post-Soviet Europe (Columns 3–4), and Central Asia (Columns 5–6). "Treated IB cities" and "Treated EB cities" report the number of unique cities within 75 km of the respective border type in each estimation sample; these need not sum to the total number of border cities reported elsewhere because a city may lie near both border types. Column (6) contains four treated external border cities and should be read as descriptive. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Coastal Access and Border Effects (Post-Soviet Europe)

	Dependent Variable: Population Growth								
	Open sea, 75 km							Incl. Caspian	Baltic & Black Sea
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
IB × Division	−0.397***	−0.240**	−0.368***					−0.402***	−0.406***
	(0.111)	(0.114)	(0.123)					(0.111)	(0.111)
EB × Division	−0.592***	−0.664***	−0.784***					−0.600***	−0.595***
	(0.201)	(0.196)	(0.227)					(0.201)	(0.200)
IB × Division × Coastal				−0.383	−0.431*	−0.482**	−0.515**		
				(0.240)	(0.242)	(0.244)	(0.246)		
IB × Division × Inland				−0.366***	−0.366***	−0.366***	−0.366***		
				(0.123)	(0.123)	(0.123)	(0.123)		
EB × Division × Coastal				0.377	0.603	0.151	0.391		
				(0.366)	(0.376)	(0.419)	(0.446)		
EB × Division × Inland				−0.782***	−0.782***	−0.782***	−0.782***		
				(0.227)	(0.227)	(0.227)	(0.227)		
Coastal × Division	−0.132	0.044		−0.282*	−0.265	−0.262	−0.250	0.101	0.068
	(0.134)	(0.142)		(0.167)	(0.167)	(0.168)	(0.168)	(0.130)	(0.134)
p: IB coastal = inland				0.951	0.812	0.672	0.588		
p: EB coastal = inland				0.007	0.002	0.051	0.019		
Coastal cities	164	164		164	161	159	156	201	136
of which near EB	23	23		23	20	19	16	30	19
Outcome mean (%)	0.583	0.583	0.610	0.583	0.584	0.583	0.584	0.583	0.583
Inland cities only			✓						
Excl. sea-only (Estonia)					✓		✓		
Excl. Kaliningrad oblast						✓	✓		
Cities	1,221	1,221	1,057	1,221	1,218	1,216	1,213	1,221	1,221
Observations	6,004	6,004	5,217	6,004	5,989	5,979	5,964	6,004	6,004
City FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Period FE	✓		✓	✓	✓	✓	✓	✓	✓
Sub-Region-by-Period FE		✓							

Notes: The dependent variable is annualized city population growth (%). "IB" and "EB" denote cities within 75 km of internal and external borders; "Division" equals one for post-1991 periods. "Coastal" equals one for cities within 75 km (straight line) of the coastline named in the column header. The open sea excludes the Caspian, a landlocked sea with no direct sea route to world markets that accounts for 37 European cities, mostly in Azerbaijan, Dagestan, and Astrakhan. Column (1) adds a coastal trajectory to the baseline specification, Column (2) replaces period fixed effects with sub-region-by-period fixed effects (Baltics, Caucasus, Slavic Europe plus Moldova), and Column (3) drops all coastal cities. Column (4) lets the border effects differ by coastal status; because the coastal main effect is included, each border coefficient compares border cities with interior cities of the same coastal status, and the p-values are from Wald tests that the coastal and inland coefficients are equal. Columns (5)–(7) re-estimate Column (4) without the three Estonian cities (Tallinn, Maardu, Paldiski) whose nearest non-FSU territory is Finland across the Gulf of Finland, without the five cities of Kaliningrad oblast, and without both; Appendix A1 lists the 23 coastal external border cities. Column (8) restores the Caspian, and Column (9) restricts the coast to the Baltic, Black, and Azov Seas, excluding the Arctic and Pacific coasts, where coastal status coincides with the withdrawal of northern and Far Eastern subsidies and with naval bases. "Coastal cities" and "of which near EB" count unique cities in each estimation sample. Central Asia is omitted because its only coastline is the Caspian. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Robustness Checks

	Dependent Variable: Population Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Europe						
IB × Division	−0.399***	−0.396***	−0.359***	−0.690***	−0.532***	−0.529***
	(0.114)	(0.110)	(0.113)	(0.121)	(0.108)	(0.118)
EB × Division	−0.587***	−0.648***	−0.442	−0.412*	−0.534***	−0.392**
	(0.204)	(0.194)	(0.291)	(0.228)	(0.180)	(0.179)
City sample	Pop 1970 <500k	Excl. 1989–99	Excl. both IB&EB	Non-conflict	Excl. Asian RU	Dist <750 km
Outcome mean (%)	0.582	0.786	0.584	0.626	0.601	0.639
Cities	1,191	1,221	1,141	1,049	1,028	771
Observations	5,856	4,783	5,629	5,193	5,039	3,754
Panel B: Central Asia						
IB × Division	0.464	0.466	0.552	0.495	0.569	0.456
	(0.376)	(0.376)	(0.395)	(0.397)	(0.378)	(0.450)
EB × Division	0.970**	0.755*	1.356**	1.442*	0.938**	0.877*
	(0.453)	(0.398)	(0.620)	(0.866)	(0.457)	(0.485)
City sample	Pop 1970 <500k	Excl. 1989–99	Excl. both IB&EB	Non-conflict	Excl. KZ–RU IB	Dist <750 km
Outcome mean (%)	1.374	1.777	1.349	1.362	1.413	1.412
Cities	172	175	170	155	165	144
Observations	791	630	785	722	755	650
City FE	✓	✓	✓	✓	✓	✓
Period FE	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is annualized city population growth (%). Column (1) excludes cities above 500,000 population in 1970, where *propiska* restrictions were most stringent. Column (2) excludes the 1989–1999 intercensal period. Column (3) excludes cities within 75 km of both internal and external borders; because these cities are dropped rather than assigned a separate indicator, its coefficients reproduce the internal-only and external-only estimates in Table A4. Column (4) drops cities near conflict-affected border pairs (Armenia–Azerbaijan, Georgia–Russia, Moldova–Ukraine, Russia–Ukraine, and Kyrgyzstan–Tajikistan for internal; Armenia–Turkey, Afghanistan–Tajikistan for external). Column (5) applies a region-specific restriction. In Panel A it excludes the 193 Russian cities in the Siberian, Far Eastern, and Urals federal districts, confining the Russian sample to districts west of the Urals and testing whether the classification of Russia with post-Soviet Europe drives the European estimates. In Panel B it excludes the ten Central Asian cities within 75 km of the Kazakhstan–Russia frontier, the one Central Asian internal border that severed a dense industrial corridor. Column (6) restricts the control group to cities within 750 km of any border, to test whether the effects are driven by contrasts with very remote interior cities. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Market Access and Urban Growth: Robustness to Spatial Autocorrelation

	Dependent Variable: Population Growth				
	(1)	(2)	(3)	(4)	(5)
Panel A: Europe					
IB × Division	−0.402***	−0.402***	−0.402***	−0.402***	−0.402**
	(0.106)	(0.122)	(0.136)	(0.146)	(0.169)
EB × Division	−0.595***	−0.595***	−0.595***	−0.595***	−0.595***
	(0.184)	(0.203)	(0.209)	(0.203)	(0.198)
Observations	6,004	6,004	6,004	6,004	6,004
Panel B: Central Asia					
IB × Division	0.471*	0.471*	0.471*	0.471*	0.471*
	(0.282)	(0.279)	(0.280)	(0.283)	(0.275)
EB × Division	0.969**	0.969**	0.969**	0.969**	0.969**
	(0.423)	(0.424)	(0.434)	(0.483)	(0.470)
Distance cutoff	25 km	50 km	75 km	100 km	200 km
Observations	805	805	805	805	805
City FE	✓	✓	✓	✓	✓
Period FE	✓	✓	✓	✓	✓

Notes: The dependent variable is annualized city population growth (%). Each column reports the baseline specification (city and period fixed effects) with Conley spatial standard errors (Conley, 1999) computed using the `acreg` package (Colella et al., 2019). Columns vary the distance cutoff beyond which spatial correlation is assumed to be zero. Point estimates are identical across columns; only standard errors change. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Distance Gradient: Border Proximity and Urban Growth

	Dependent Variable: Population Growth			
	Europe		Central Asia	
	(1)	(2)	(3)	(4)
IB (0–25 km) × Division	−0.486***	−0.264	0.633	0.886**
	(0.170)	(0.171)	(0.434)	(0.436)
IB (25–50 km) × Division	−0.363**	−0.195	0.488	0.768
	(0.155)	(0.149)	(0.614)	(0.631)
IB (50–75 km) × Division	−0.225	−0.118	0.045	0.374
	(0.196)	(0.197)	(0.574)	(0.579)
IB (75–100 km) × Division	0.105	0.184	−0.233	0.200
	(0.168)	(0.167)	(0.475)	(0.485)
EB (0–25 km) × Division	−1.055***	−1.224***	1.438**	1.362**
	(0.360)	(0.372)	(0.626)	(0.630)
EB (25–50 km) × Division	−0.377	−0.447	0.148	0.440
	(0.384)	(0.363)	(0.391)	(0.392)
EB (50–75 km) × Division	−0.485*	−0.561**	0.755	0.084
	(0.258)	(0.254)	(0.636)	(0.337)
EB (75–100 km) × Division	−0.835***	−0.925***	0.953	0.257
	(0.206)	(0.224)	(1.126)	(0.982)
Outcome mean (%)	0.583	0.583	1.369	1.309
Cities	1,221	1,221	175	166
Observations	6,004	6,004	805	769
City FE	✓	✓	✓	✓
Period FE	✓		✓	✓
Sub-Region-by-Period FE		✓		

Notes: The dependent variable is annualized city population growth (%). Each row reports the coefficient on the interaction between a distance-bin indicator and a post-1991 indicator. The omitted reference category is cities beyond 100 km from any border. Column (1) is the baseline for Europe with city and period fixed effects. Column (2) replaces period fixed effects with sub-region-by-period fixed effects (Baltics, Caucasus, Slavic Europe plus Moldova). Column (3) is the baseline for Central Asia. Column (4) excludes Turkmenistan. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A10: Alternative Distance Thresholds for Border Proximity

	Dependent Variable: Population Growth				
	(1)	(2)	(3)	(4)	(5)
Distance threshold	25 km	50 km	75 km	100 km	125 km
Panel A: Europe					
IB × Division	−0.537***	−0.442***	−0.402***	−0.255**	−0.178*
	(0.167)	(0.121)	(0.111)	(0.105)	(0.102)
EB × Division	−1.051***	−0.676**	−0.595***	−0.713***	−0.833***
	(0.353)	(0.267)	(0.200)	(0.168)	(0.149)
Treated IB cities	107	246	353	446	528
Treated EB cities	47	94	142	185	225
Interior cities	1,074	921	806	709	626
Outcome mean (%)			0.583		
Cities			1,221		
Observations			6,004		
Panel B: Central Asia					
IB × Division	0.457	0.544	0.471	0.427	0.457
	(0.319)	(0.347)	(0.370)	(0.403)	(0.442)
EB × Division	1.374**	1.123**	0.969**	0.928**	0.542
	(0.590)	(0.525)	(0.451)	(0.457)	(0.397)
Treated IB cities	58	88	105	114	124
Treated EB cities	9	11	15	19	25
Interior cities	108	78	60	49	40
Outcome mean (%)			1.369		
Cities			175		
Observations			805		
City FE			✓		
Period FE			✓		

Notes: The dependent variable is annualized city population growth (%). Each column redefines both treatments at the stated distance threshold: a city is treated if it lies within that distance of the relevant frontier, and the control group is cities beyond it. Column (3) reproduces the baseline 75 km specification. The estimation sample is held fixed across columns, so only the allocation of cities between treated and control groups changes; "Treated IB" and "Treated EB" overlap for cities near both frontiers, so the three counts do not sum to the sample total. Internal border effects attenuate monotonically as the threshold widens, which is what a frontier-specific shock implies, since wider rings add cities with weaker exposure. External border effects do not, consistent with the broader peripherality component discussed in the text and in Table A9. In Panel B the interior control group falls from 108 cities at 25 km to 40 at 125 km, so the widest Central Asian columns rest on a thin comparison group. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Summary Statistics: Sectoral Composition (1989)

	IB Regression Sample					EB Regression Sample				
	Mean	Med.	SD	Min	Max	Mean	Med.	SD	Min	Max
Panel A: Europe										
Manufacturing share	0.834	0.925	0.234	0	1	0.841	0.924	0.224	0	1
Agriculture share	0.026	0	0.088	0	1	0.022	0	0.068	0	0.813
Mining share	0.074	0	0.204	0	1	0.069	0	0.196	0	1
Total empl. (000s)	17.3	5.381	40.6	0.022	785	17.2	5.653	41.5	0.022	785
(total)	1021					848				
of which: IB cities	290 (0.284)					62 (0.073)				
of which: EB cities	62 (0.061)					117 (0.138)				
of which: Interior	731 (0.716)					731 (0.862)				
Heterogeneity diagnostics (treated cities):										
Manuf. quartile (Q1/Q2/Q3/Q4)	249(77)/258(59)/255(64)/259(90)					203(31)/221(22)/220(29)/204(35)				
Manuf. median split (High/Low)	514(154) / 507(136)					424(64) / 424(53)				
Agric. median split (High/Low)	494(127) / 527(163)					414(47) / 434(70)				
Mining median split (High/Low)	294(90) / 727(200)					240(36) / 608(81)				
Panel B: Central Asia										
Manufacturing share	0.753	0.849	0.281	0	1	0.686	0.755	0.286	0	1
Agriculture share	0.036	0	0.111	0	1	0.074	0.001	0.17	0	1
Mining share	0.111	0	0.253	0	1	0.102	0	0.225	0	0.93
Total empl. (000s)	12.1	4.728	20.5	0.026	143	10.1	3.03	18.3	0.118	118
(total)	141					61				
of which: IB cities	91 (0.645)					3 (0.049)				
of which: EB cities	3 (0.021)					11 (0.18)				
of which: Interior	50 (0.355)					50 (0.82)				
Heterogeneity diagnostics (treated cities):										
Manuf. median split (High/Low)	72(55) / 69(36)					19(2) / 42(9)				
Agric. median split (High/Low)	69(45) / 72(46)					32(8) / 29(3)				
Mining median split (High/Low)	45(27) / 96(64)					19(1) / 42(10)				

Notes: Sectoral shares are calculated as sector employment divided by total city employment in 1989. The IB regression sample includes all cities except those near only external borders, matching internal border specifications (Tables 1–2). The EB regression sample includes all cities except those near only internal borders. "Of which" rows report the count and share of each city type; IB and EB counts need not sum to the total because some cities fall within 75 km of both border types. In the heterogeneity diagnostics, numbers in parentheses indicate treated (border) cities; the remainder are interior controls. "Manuf. quartile" reports the Q1/Q2/Q3/Q4 distribution across manufacturing quartiles used in Table 2, Columns (1)–(2). "Median split (High/Low)" reports cell sizes for specifications in Tables 2 and A12. In both regions, the median agriculture and mining shares are zero, so "High" captures cities with any positive employment in that sector.

Table A12: Urban Growth and Sectoral Composition: Additional Sector Splits

	Dependent Variable: Population Growth			
	Internal Border		External Border	
	(1)	(2)	(3)	(4)
	Panel A: Europe			
	Agric.	Mining	Agric.	Mining
IB × Division × Any agric.	−0.307*			
	(0.165)			
IB × Division × No agric.	−0.446**			
	(0.181)			
IB × Division × Any mining		0.059		
		(0.196)		
IB × Division × No mining		−0.578***		
		(0.155)		
EB × Division × Any agric.			−0.584	
			(0.387)	
EB × Division × No agric.			−0.758**	
			(0.295)	
EB × Division × Any mining				−0.170
				(0.314)
EB × Division × No mining				−0.915***
				(0.305)
Any agric. × Division	0.005		0.005	
	(0.126)		(0.126)	
Any mining × Division		−0.215		−0.215
		(0.146)		(0.146)
p-value: equal effects	0.570	0.011	0.720	0.090
Outcome mean (%)	0.527	0.527	0.578	0.578
Cities	1,021	1,021	848	848
Observations	5,033	5,033	4,206	4,206
	Panel B: Central Asia			
	Manuf.	Mining	Manuf.	Mining
IB × Division × High manuf.	0.197			
	(0.529)			
IB × Division × Low manuf.	0.499			
	(0.604)			
IB × Division × Any mining		1.470*		
		(0.859)		
IB × Division × No mining		−0.133		
		(0.445)		
EB × Division × High manuf.			1.483	
			(1.291)	
EB × Division × Low manuf.			1.479*	
			(0.778)	
EB × Division × Any mining				1.702**
				(0.840)
EB × Division × No mining				0.915
				(0.652)
High manuf. × Division	0.476		0.475	
	(0.711)		(0.717)	
Any mining × Division		−1.577*		−1.577*
		(0.894)		(0.902)
p-value: equal effects	0.705	0.099	0.997	0.453
Outcome mean (%)	1.362	1.362	1.328	1.328
Cities	141	141	61	61
Observations	652	652	284	284
City FE	✓	✓	✓	✓
Period FE	✓	✓	✓	✓

Notes: The dependent variable is annualized city population growth (%). IB and EB denote cities within 75 km of an internal and an external border; Division equals one for post-1991 periods. Each column estimates the border effect separately for the two groups of cities defined by the indicated 1989 sector and includes the group indicator interacted with Division. For agriculture and mining, the within-region median employment share is zero, so "Any" denotes cities with positive employment in the sector and "No" those without; Central Asian manufacturing is split at its median share of 0.848, with "High" denoting cities above the median. Internal-border columns exclude external-only cities and external-border columns exclude internal-only cities, so each compares one border type with interior cities. Manufacturing quartiles for Europe and the agriculture split for Central Asia appear in Table 2; utilities and repair appear in Table A13. Of the 314 European cities with mining employment, 159 are in the lowest manufacturing quartile. The Central Asian external-border sample contains 61 cities, 11 of them near external borders, so Panel B, Columns (3)–(4), should be read with caution. Standard errors clustered at the city level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A13: Urban Growth and Sectoral Composition: Utilities and Repair in Post-Soviet Europe

	Dependent Variable: Population Growth			
	Internal Border		External Border	
	(1) Utilities	(2) Repair	(3) Utilities	(4) Repair
IB × Division × Any utilities	−0.131			
	(0.224)			
IB × Division × No utilities	−0.585***			
	(0.147)			
IB × Division × Any repair		−0.413**		
		(0.187)		
IB × Division × No repair		−0.368**		
		(0.156)		
EB × Division × Any utilities			−0.018	
			(0.426)	
EB × Division × No utilities			−1.088***	
			(0.275)	
EB × Division × Any repair				−0.374
				(0.407)
EB × Division × No repair				−0.851***
				(0.282)
Any utilities × Division	−0.556***		−0.556***	
	(0.129)		(0.129)	
Any repair × Division		0.070		0.070
		(0.134)		(0.134)
p-value: equal effects	0.090	0.854	0.035	0.336
Outcome mean (%)	0.527	0.527	0.578	0.578
Cities	1,021	1,021	848	848
Observations	5,033	5,033	4,206	4,206
City FE	✓	✓	✓	✓
Period FE	✓	✓	✓	✓

Notes: The dependent variable is annualized city population growth (%). IB and EB denote cities within 75 km of an internal and an external border; Division equals one for post-1991 periods. Each column estimates the border effect separately for European cities with and without 1989 employment in the indicated sector and includes that indicator interacted with Division. The within-region median employment share is zero for both sectors, so "Any" denotes cities with positive employment in the sector and "No" those without. Utilities comprise electric power, gas transmission, district heating, and water supply, plus a small number of water-transport records; repair comprises industrial repair and maintenance. Utilities sold much of their output locally but relied on fuel and network inputs that crossed republican boundaries, so these splits are reported for completeness rather than as tests of the mechanism. Internal-border columns exclude external-only cities and external-border columns exclude internal-only cities, so each compares one border type with interior cities. Central Asian splits for these sectors are not reported because their external-border cells contain at most six treated cities. Standard errors clustered at the city level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A14: Covariate Balance in Matched Samples

	Treated (Border)					Control (Interior)				
Matching variable	Mean	SD	Min	Max	*N*	Mean	SD	Min	Max	*N*
Panel A: Europe										
Internal border matching:										
1970 Population	61,465	136,889	5,235	1,222,852	352	70,598	140,246	5,226	1,170,133	415
1989 Ind. Employment	15,564	31,924	54.0	250,404	289	15,634	30,281	24.0	274,489	399
1989 Ind. Turnover	340,085	644,501	70.0	4,758,280	290	370,136	696,305	64.0	4,796,282	379
1989 Manufacturing	216,068	463,404	0.000	3,781,983	290	259,122	514,307	0.000	3,728,328	383
1989 SIC	13,427	6,547	2,730	32,720	279	12,564	6,432	2,730	32,720	719
Military count	8.084	17.8	1.000	160.0	179	18.6	29.5	1.000	208.0	94
Military size	17.6	38.0	2.000	339.0	179	40.6	63.0	2.000	443.0	94
External border matching:										
1970 Population	49,433	104,595	5,235	766,705	141	53,785	108,825	5,226	770,905	238
1989 Ind. Employment	12,137	24,526	54.0	161,117	115	13,034	25,561	59.0	158,143	199
1989 Ind. Turnover	256,859	504,018	70.0	3,176,533	117	254,788	505,249	64.0	3,107,729	207
1989 Manufacturing	159,088	326,866	0.000	2,117,026	117	168,095	339,349	0.000	2,161,119	198
1989 SIC	13,673	6,316	2,730	27,520	115	12,196	6,151	2,730	27,520	826
Military count	7.758	15.4	1.000	105.0	66	14.7	28.1	1.000	166.0	49
Military size	16.5	32.5	2.000	218.0	66	31.7	60.1	2.000	360.0	49
Panel B: Central Asia										
Internal border matching:										
1970 Population	54,241	76,263	6,197	430,618	97	57,006	81,603	6,010	523,271	62
1989 Ind. Employment	11,542	16,064	129.0	87,890	86	11,075	19,623	118.0	117,556	52
1989 Ind. Turnover	312,309	448,680	1,330	2,403,345	88	456,404	672,555	954.0	2,731,450	48
1989 Manufacturing	191,509	340,011	0.000	1,825,853	91	187,719	319,809	0.000	1,414,383	49
1989 SIC	10,820	6,123	2,730	23,000	69	11,694	6,408	2,730	23,000	42
Military count	3.209	4.877	1.000	28.0	43	3.200	2.484	1.000	10.0	15
Military size	7.047	10.2	2.000	58.0	43	7.267	5.788	2.000	23.0	15
External border matching:										
1970 Population	34,647	63,901	6,010	253,118	14	41,022	72,711	5,879	266,815	21
1989 Ind. Employment	3,771	6,317	163.0	21,598	11	4,166	6,473	147.0	22,022	20
1989 Ind. Turnover	98,293	130,617	2,523	382,986	11	104,737	131,277	2,080	386,449	20
1989 Manufacturing	51,211	86,545	0.000	281,351	11	57,188	88,302	0.000	302,282	20
1989 SIC	8,169	4,720	2,730	20,110	9	9,170	5,908	2,730	20,110	48
Military count	1.333	0.577	1.000	2.000	3	1.500	0.577	1.000	2.000	4
Military size	2.667	1.155	2.000	4.000	3	3.000	1.155	2.000	4.000	4

Notes: Each row reports summary statistics of the matching variable for treated (border) and control (interior) cities in the corresponding matched sample. Each matching exercise is independent: the row for "1970 Population" under internal border matching reports balance in the sample matched on 1970 population (Table 3, Column 1 for Europe; Table A16, Column 1 for Central Asia). Columns (1)–(4) and (6) of the matching tables use 1:2 nearest-neighbor propensity score matching with replacement on the city cross-section. The 1:2 ratio balances two considerations: using more than one neighbor improves efficiency in the small Central Asian samples, while restricting to two avoids drawing poor matches from distant regions of the covariate distribution. With replacement and a fixed neighbor count, exact balance on the matching variable is not guaranteed, particularly when the treated and control distributions have different shapes or when a few control cities serve as neighbors for many treated cities. Column (5) uses exact matching on each city's primary SIC code, retaining all border and interior cities that shared the same dominant industrial classification in 1989. Military matching uses two covariates (count and workforce size of defense enterprises closed 1989–1991), so both are reported. Statistics are computed on the cross-section (one observation per city).

Table A15: Robustness to Matching on Pre-1991 Characteristics (Central Asia)

	Dependent Variable: Population Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Internal Borders						
IB × Division	0.597	0.511	0.607	0.733	0.026	1.424*
	(0.424)	(0.481)	(0.507)	(0.468)	(0.408)	(0.711)
Matching on	1970 Pop.	1989 Ind. Empl.	1989 Ind. Turn.	1989 Manuf.	1989 SIC	Military
Outcome mean (%)	1.332	1.344	1.382	1.389	1.548	1.593
Cities	151	131	129	133	106	58
Observations	697	603	595	614	483	266
Adj. R^2	0.227	0.289	0.291	0.328	0.292	0.455
Panel B: External Borders						
EB × Division	3.007*	1.086	0.736	2.699	−0.533	3.819*
	(1.696)	(1.040)	(1.113)	(2.139)	(0.651)	(1.450)
Matching on	1970 Pop.	1989 Ind. Empl.	1989 Ind. Turn.	1989 Manuf.	1989 SIC	Military
Outcome mean (%)	1.525	1.553	1.515	1.486	1.746	2.294
Cities	21	16	23	19	25	5
Observations	91	69	102	83	110	22
Adj. R^2	0.229	0.192	0.083	0.012	0.306	0.451
City FE	✓	✓	✓	✓	✓	✓
Period FE	✓	✓	✓	✓	✓	✓

Notes: Each column re-estimates the baseline specification on a matched subsample of treated border cities and their nearest neighbors. Columns (1)–(4) and (6) use 1:2 nearest-neighbor propensity score matching with replacement on the city cross-section. Column (5) uses exact matching on each city's primary SIC code, restricting the sample to border and interior cities that shared the same dominant industrial classification in 1989. In Central Asia, matched samples are small, particularly for external borders (Panel B), where the treated sample contains only 11–15 cities depending on covariate availability. The negative EB coefficient in Column (5) of Panel B (−0.533) rests on just 25 cities and likely reflects the composition of the few SIC strata that contain both treated and control cities rather than a substantive finding. Column (6) of Panel B retains just 5 cities, so this estimate should not be over-interpreted. "Military" matches jointly on the count and total workforce of defense enterprises that closed between 1989 and 1991. The dependent variable is annualized city population growth (%). "Division" equals one for post-1991 periods. All specifications include city and period fixed effects. Standard errors clustered at the city level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A16: Matching Design Variations (Europe)

	Dependent Variable: Population Growth					
	Internal Borders			External Borders		
	(1)	(2)	(3)	(4)	(5)	(6)
IB × Division	−0.505***	−0.600***	−0.392***			
	(0.132)	(0.131)	(0.138)			
EB × Division				−0.802***	−0.699***	−0.526*
				(0.262)	(0.258)	(0.267)
Control radius	None	≤1,000 km	≤500 km	None	≤1,000 km	≤500 km
Median pair dist. (km)	1,326	637	347	—	636	334
Treated cities matched	290	290	290	117	117	117
Outcome mean (%)	0.464	0.515	0.531	0.606	0.626	0.677
Cities	626	588	520	260	217	200
Observations	3,062	2,870	2,529	1,272	1,058	973
City FE	✓	✓	✓	✓	✓	✓
Period FE	✓	✓	✓	✓	✓	✓

Notes: Each column re-estimates the baseline specification on a matched subsample. All columns condition on all six pre-1991 covariates simultaneously: 1970 population, 1989 industrial employment, turnover, and manufacturing output, and the count and workforce of defence enterprises closing between 1989 and 1991. Matching is 1:2 nearest-neighbour with replacement under common support, with the propensity score estimated by probit on the city cross-section of the 1,076 post-Soviet European cities with complete covariate data. Columns (1) and (4) impose no geographic restriction. Columns (2)–(3) and (5)–(6) additionally require each matched control to lie within the stated great-circle distance of its treated city, following Redding and Sturm (2008), who impose a geographic band on their control group; because `psmatch2` offers no caliper on an auxiliary variable, the nearest-neighbour selection in these columns is performed manually on the estimated score. Every treated city retains two matched controls at each radius, except that two external-border cities retain one at 500 km. The median unconstrained pair distance for external borders is not reported because that match is reported in Column (4) only. "Cities" reports unique cities in the matched sample. The dependent variable is annualized city population growth (%). "Division" equals one for post-1991 periods. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A17: Excluding Science Cities and Closed Cities

	Dependent Variable: Population Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
IB × Division	−0.395***	−0.392***	−0.379***	−0.365***	−0.361***	0.479
	(0.110)	(0.110)	(0.111)	(0.112)	(0.113)	(0.384)
EB × Division	−0.609***	−0.607***	−0.600***	−0.592***	−0.593***	0.833*
	(0.200)	(0.200)	(0.200)	(0.201)	(0.202)	(0.459)
Outcome mean (%)	0.553	0.554	0.547	0.540	0.543	1.355
Sample	Europe	Europe	Europe	Europe	Europe	CA
Science cities excl.	3 km	5 km	10 km	20 km	20 km	
Closed cities excl.					✓	✓
Cities	1,170	1,166	1,150	1,119	1,112	162
Observations	5,749	5,729	5,649	5,494	5,460	745
City FE	✓	✓	✓	✓	✓	✓
Period FE	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is annualized city population growth (%). "IB" and "EB" denote cities within 75 km of internal and external borders, respectively; "Division" equals one for post-1991 periods. Each column re-estimates the baseline specification after excluding Soviet Science Cities, other closed settlements, or both. These places lay mostly in the interior, and because Science Cities outperformed comparable localities after 1991 (Schweiger et al., 2022), as controls they could inflate the estimated border penalties. Columns (1)–(4) exclude sample cities within 3, 5, 10, and 20 km of the 91 of the 95 Science Cities in Schweiger et al. (2022) that can be located; the list covers modern Russia only, so these columns bear on the European sample. Matching is by coordinates rather than names, because Schweiger et al. (2022) define Science Cities at the level of the Russian municipality, many places were renamed after 1991, and statistics were never published for closed cities; roughly a third of Science Cities were built from scratch and fall below the sample's 1970 population threshold. The exclusion removes 51 sample cities at 3 km and 102 at 20 km, since one Science City can neighbor several sample towns; beyond 10 km the additional exclusions are adjacent towns rather than further Science Cities, so Column (4) is best read as a neighborhood exclusion. Only one matched Science City lies within 75 km of an internal border. Column (5) additionally excludes 7 closed or restricted-access cities in post-Soviet Europe outside Russia (Paldiski, Sillamäe, Liepāja, Sevastopol, Zhovti Vody, Kamianske, and Dnipro), and Column (6) excludes 13 in Central Asia (Baikonur, Stepnogorsk, Aktau, Ust-Kamenogorsk, Chkalovsk, Taboshar, Khaidarkan, Kadamjay, Kara-Balta, Mailuu-Suu, Zarafshon, Chirchik, and Navoiy), all hand-coded from secondary sources. Because closed cities were largely omitted from the published Soviet statistics on which the population panel is built, several of the most restricted sites (Sarov, Ozyorsk, Seversk, Novouralsk, Lesnoy, Kurchatov, Sniečkus, and Slavutych) do not appear in the sample at all. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix A1 - Data Cleaning

Defence Industry Data Matching

The raw database contains over 33,000 enterprise records spanning the full Soviet period. I restrict the sample to enterprises whose military work ended between 1989 and 1991, the window surrounding dissolution, and whose workforce size is documented (categories 1 through 3 on the database's ordinal scale). Enterprises that ceased military production before 1970 or began after 1991 are excluded. Because city names in the defense database do not map directly to the population panel, I match locations using a combination of automated string matching and manual resolution, successfully linking 822 in the filtered database to cities in the main sample. For each matched city, I construct two variables: the count of defense enterprises that stopped work in the 1989–1991 window, and the sum of their ordinal workforce sizes.

Night Lights Data

Several steps were undertaken to transform pixel-level lights data into city-level data following Storeygard (2016). First, 30 satellite-years of night lights data were merged into a single binary grid, indicating whether a pixel was lit above a digital number threshold of 20 in at least one year.[12] These ever-lit areas were then converted into polygons by aggregating contiguous lit pixels. For each polygon and year, I summed the raw digital number values (ranging from 0 to 63) of all constituent pixels to obtain a continuous measure of total light intensity. The regression outcome is the log of this summed intensity, which captures variation in economic activity across cities and over time.

I matched polygons to known cities using census population data with latitude and longitude coordinates from Brinkhoff (2021). Polygons that did not correspond to a known city were excluded; these likely represent forest fires, sensor noise not flagged by the satellite algorithm, or smaller towns and large villages, and account for 10 to 20 percent of total digital numbers in the 15-country sample.[13]

Additionally, I removed lights from gas flares, as identified by Elvidge et al. (2009). To further minimize errors, I clipped the nighttime lights raster to the national boundaries of FSU countries. Without this step, light emitted by cities in neighboring non-FSU countries (e.g., Poland, Finland, China) could bleed into FSU border areas through the well-documented blooming effect in DMSP data, artificially inflating light intensity for FSU cities near external borders. The night lights dataset is annual from 1992 to 2013, enabling analysis only for the period following the dissolution.[14]

[12] The threshold of 20 is applied only at the polygon construction stage to define city boundaries by identifying reliably lit urban areas; it does not affect the outcome variable used in regressions. Li et al. (2020) show that thresholds of 20 to 30 effectively eliminate blooming artifacts and sensor noise around urban peripheries while retaining reliably lit areas.

[13] An alternative approach would use all lit pixels within a fixed buffer around each city centroid, without a brightness threshold. This would avoid potential censoring of newly developed areas at urban peripheries but would also introduce noise from non-urban light sources. The polygon-based approach used here ties the nightlights outcome directly to known urban footprints identified in the census data, ensuring that the measured light corresponds to the same cities whose population growth is analyzed in the main results.

[14] Gibson et al. (2021) highlight key issues with DMSP night lights data, such as blurring and top-coding, which can distort economic measurements. They particularly note that these errors lead to poor predictions of economic activity in low-density rural areas. However, the study focuses on whole cities with populations of at least five thousand people in 1970, where these data provide more reliable insights. Additionally, the results are robust to dropping large cities such as Moscow, where top-coding may be an issue.

Coastal External Border Cities

Twenty-three post-Soviet European cities lie within 75 km of both an external border and the open sea (Table A6, Column 4). They are listed below by country, with the nearest non-FSU country in parentheses; † marks cities also within 75 km of an internal border.

- *Estonia* (Finland, across the Gulf of Finland): Maardu, Paldiski, Tallinn.
- *Georgia* (Turkey): Batumi, Chakvi, Kobuleti, Ozurgeti, Poti, Samtredia.
- *Moldova* (Romania): Ceadîr-Lunga†, Taraclia†.
- *Russia, Kaliningrad oblast* (Poland): Baltiysk, Chernyakhovsk†, Gusev†, Kaliningrad†.
- *Russia, elsewhere*: Kovdor and Vyborg (Finland); Artyom, Ussuriysk, and Vladivostok (China).
- *Ukraine* (Romania): Izmail†, Kiliya†, Reni†.

Coastal status is defined by distance (75 km), so the group includes towns well inland, among them the two Moldovan towns, whose nearest coast lies in Ukraine.

Appendix B - Effects by Individual Border Pair

To further probe how the effect differs for each border pair, I estimate the following three-way interaction model separately for internal and external borders:

$$y_{ct} = \sum_{j=1}^{J} \beta_j \big(\text{border}_c \times \text{division}_t \times \mathbf{1}\{\text{borderpair}_c = j\}\big) + \alpha_c + d_t + \epsilon_{ct}, \tag{7}$$

where border_c indicates proximity to internal or external borders, $\mathbf{1}\{\text{borderpair}_c = j\}$ is an indicator equal to one if city c's nearest border belongs to border pair j and zero otherwise, and J is the total number of border pairs. City fixed effects α_c absorb all time-invariant city characteristics, including border-pair membership, while year fixed effects d_t control for common census year shocks. This model allows isolation of how the Soviet Union's dissolution uniquely influenced population growth around distinct borders.

Figure B1 presents results for internal border pairs. In Europe, the majority of border pairs exhibit negative coefficients, though magnitudes and precision vary considerably with sample size. The largest and most precisely estimated declines are at Latvia–Lithuania (-1.70, significant at the 1% level, 31 cities), Moldova–Ukraine (-1.61, significant at the 1% level, 46 cities), Belarus–Lithuania (-1.43, significant at the 1% level, 22 cities), and Belarus–Ukraine (-1.07, significant at the 1% level, 26 cities). Armenia–Azerbaijan shows a large negative effect (-1.76, significant at the 1% level), though the Nagorno-Karabakh conflict complicates interpretation. Belarus–Russia (-0.35, significant at the 5% level) and Lithuania–Russia (-1.09, significant at the 1% level) are also negative, while Russia–Ukraine is modestly positive (0.30, significant at the 5% level), potentially reflecting the continued dominance of bilateral Russia–Ukraine trade despite border frictions. This uplift requires specific context: rather than cleanly severing ties, the economies of eastern Ukraine and western Russia remained deeply intertwined throughout the 1990s, particularly in heavy industry, metallurgy, and energy transit. The imposition of a national boundary transformed these previously integrated neighboring cities into essential customs, logistics, and transit hubs for the largest bilateral trade corridor in the former Soviet Union. Several pairs with very few cities (Belarus–Latvia with 1 city, Azerbaijan–Russia with 7 cities, Latvia–Russia with 4 cities) produce extreme or imprecise estimates that should not be over-interpreted.

In Central Asia, six of the seven internal border pairs show positive coefficients, with Kazakhstan–Russia the exception. Tajikistan–Uzbekistan (1.05, $p = 0.09$) and Turkmenistan–Uzbekistan (1.08, $p = 0.10$) are the most precisely estimated, though the intervals remain wide throughout. The pair-level estimates are generally imprecise, with at most weak evidence of positive internal-border responses. This pattern is consistent with the positive but statistically insignificant full-sample Central Asian internal coefficient in Table 1, Column (4).

Figure B2 presents results for external border pairs. In Europe, the largest and most precisely estimated negative effects are at Moldova–Romania (-1.78, significant at the 1% level, 26 cities) and Lithuania–Poland (-1.97, significant at the 1% level, 9 cities), both on the western frontier; all nine Lithuanian towns also lie within 75 km of an internal frontier, so their decline compounds the two shocks (Table A4). Belarus–Poland is also negative (-0.58, significant at the 5% level). Several Caucasian border pairs show mixed results: Armenia–Turkey (-0.58) and Georgia–Turkey (-0.35) are negative but imprecise, consistent with the limited commercial potential of these frontiers noted in Section 2. Poland–Russia shows a notable positive effect (1.01, significant at the 1% level), likely reflecting Kaliningrad's unique economic arrangements as a Russian exclave with special trade provisions. Border pairs with very few cities (Hungary–Ukraine with 2 cities, Finland–Russia with 4 cities) are imprecise.

In Central Asia, all external border pairs show positive coefficients. Iran–Turkmenistan has the largest effect (3.68, significant at the 1% level), followed by Afghanistan–Tajikistan

(1.55, significant at the 5% level). China–Kazakhstan (0.68, $p = 0.13$) is positive but imprecise, reflecting the single city (Zharkent) near this border. The uniformly positive direction across geographically distinct frontiers reinforces the interpretation that border opening generated market access gains in Central Asia. To reiterate, individual estimates are imprecise for pairs with few cities, so the value of this exercise lies in the broad pattern rather than any single coefficient.

Figure B2 also reports Russia's Asian frontiers, shown separately under "Other" because they face different neighbours from either regional group. These cities are external border cities on the same 75 km rule as all others and are estimated within the European regression, since Russia is classified with post-Soviet Europe throughout (Figure 1). The China–Russia estimate is -1.524 on 11 cities ($p = 0.23$) and Mongolia–Russia is 0.847 on 2 cities ($p = 0.15$). Neither is distinguishable from zero and the two carry opposite signs, so the pair-level evidence does not identify an effect on either frontier. The China–Russia estimate does not provide evidence of a comparable positive effect, but does not establish a difference from Central Asia. The Central Asian estimate pools frontiers with Afghanistan and Iran alongside China, and the China–Kazakhstan cell contains a single city, so the regional gain cannot be attributed specifically to Chinese market access.

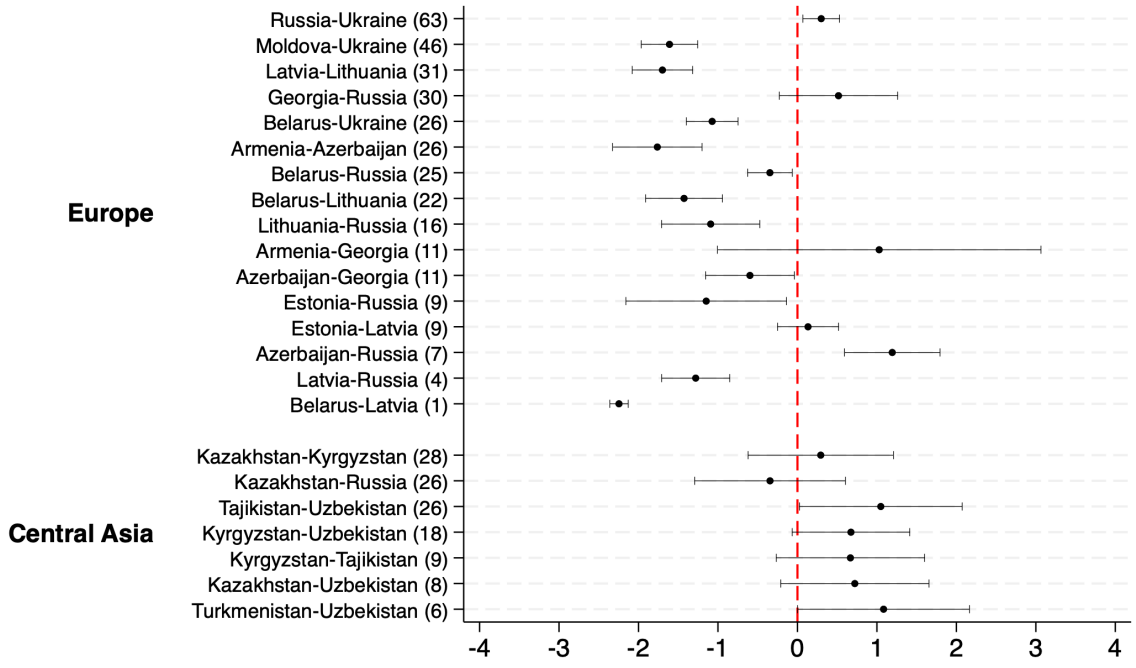


Figure B1: Treatment Effects by Internal Border Pair

Notes: Each coefficient represents the post-1991 effect on annualized population growth for cities within 75 km of the indicated internal border pair, estimated from equation 7. Separate regressions are run for Europe and Central Asia, with interior cities in each region as controls. Numbers in parentheses indicate the number of border cities per pair. Border pairs sorted by city count within each region. All specifications include city, period, and border-pair fixed effects. Standard errors clustered at the city level. Bars show 90% confidence intervals.

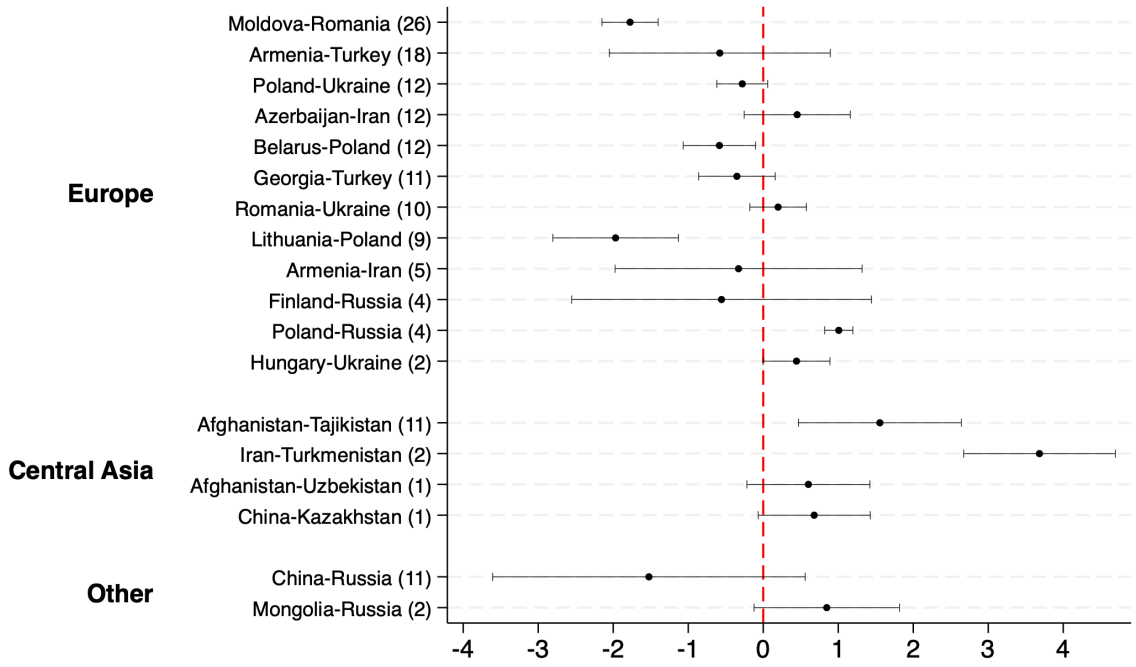


Figure B2: Treatment Effects by External Border Pair

Notes: Each coefficient represents the post-1991 effect on annualized population growth for cities within 75 km of the indicated external border pair, estimated from equation 7. Two regressions are run, one for post-Soviet Europe and one for Central Asia, with interior cities in each region as controls. The groupings are for display only: pairs labelled "Other" are Russia's Asian frontiers and are estimated within the European regression, since Russia is classified with post-Soviet Europe throughout (Figure 1). Numbers in parentheses indicate the number of treated border cities per pair. Two pairs are estimated but omitted from the plot: Slovakia–Ukraine, which contains one city and is collinear, and Estonia–Finland, where the three cities reach Finland across the Gulf of Finland rather than over a land frontier. All specifications include city, period, and border-pair fixed effects. Standard errors clustered at the city level. Bars show 90% confidence intervals.

Appendix C - Effects on Economic Activity Proxied by Night Lights

In this section, I examine whether the city-level population results for Europe and Central Asia also manifest in nighttime lights data, a widely used proxy for local economic activity. Although nighttime light data is only available from 1992 onward, I exploit this limitation by treating 1992, the first full year after the Soviet dissolution, as the omitted baseline. I estimate separate event-study specifications for internal and external border cities, comparing changes in log night lights relative to interior cities over the period 1993–2013.

$$\log(\text{lights})_{ct} = \sum_{\tau=1993}^{2013} \beta_\tau \big(\text{border}_c \times \mathbf{1}\{t = \tau\}\big) + \alpha_c + d_t + \epsilon_{ct}, \tag{8}$$

where $\log(\text{lights})_{ct}$ is the log of summed lights in city c in year t, border_c is a dummy equal to one if a city is within 75 km of a border and zero otherwise, and $\mathbf{1}\{t = \tau\}$ is an indicator for year τ. The year 1992 serves as the omitted baseline, so each coefficient β_τ captures the difference in log night lights between border and interior cities in year τ relative to 1992. City fixed effects α_c absorb time-invariant city characteristics, while year fixed effects d_t control for common annual shocks. The specification is estimated separately for internal border cities (IB) and external border cities (EB), with the two sets of coefficients plotted together in Figure C1. While this approach does not allow for direct observation of pre-1992 trends, it reveals important information about the relative trajectories of economic activity following the dissolution.

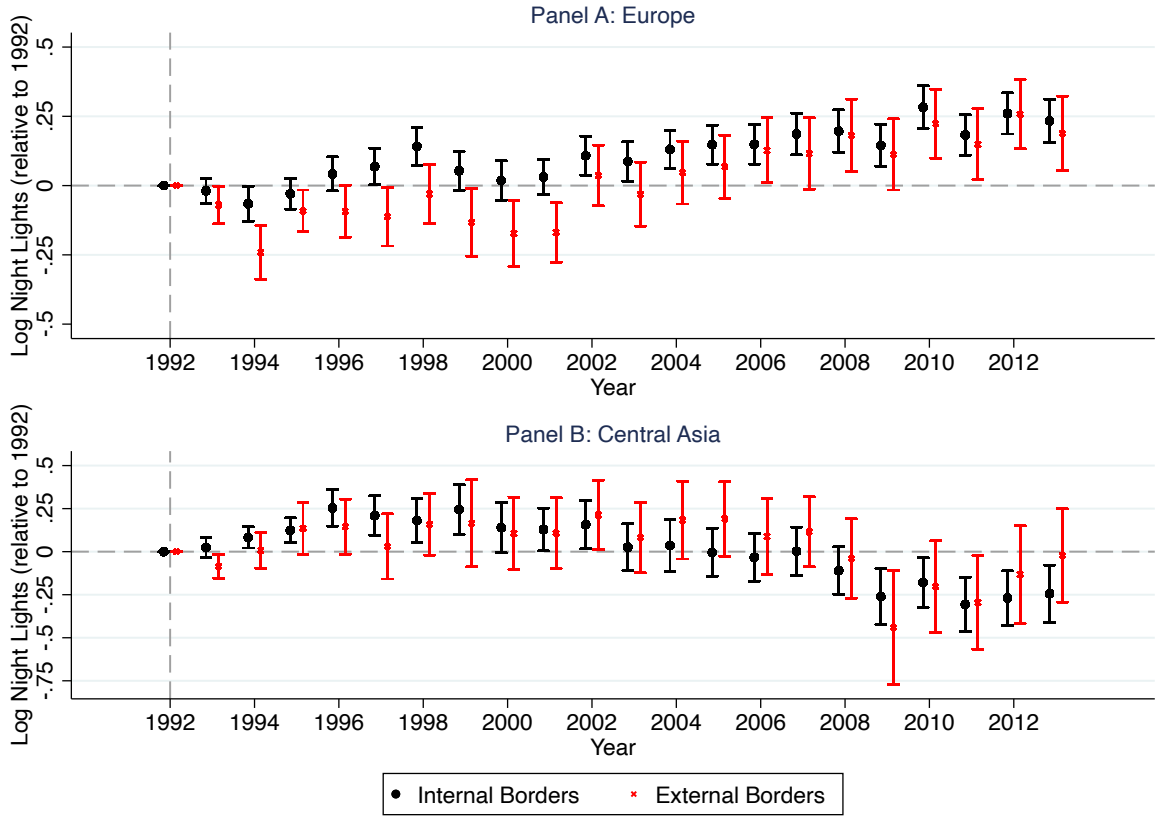


Figure C1: Event-study Estimates of Nighttime Light Intensity relative to 1992

Notes: This figure plots coefficients from event-study regressions of log nighttime light intensity on border proximity interacted with year indicators, relative to the 1992 baseline. Black circles denote internal borders; red X's denote external borders. Panel A shows Europe; Panel B shows Central Asia. Because nightlights data begin in 1992, no pre-treatment trends can be assessed. All regressions include city, year, and border-pair fixed effects. Standard errors clustered at the city level; bars show 90% confidence intervals.

Figure C1 presents results for Europe (Panel A) and Central Asia (Panel B). Two important caveats apply when interpreting these results alongside the main population findings. First, nighttime lights data begin in 1992, so there is no pre-treatment period to validate parallel trends. Second, the population regressions estimate annualized growth rates (a flow measure), while the nightlights specification estimates log levels relative to a 1992 baseline. A negative coefficient in the population regression implies continuously diverging population levels, whereas the nightlights coefficients capture the cumulative deviation in economic activity relative to 1992. These differences in outcome measurement mean the two sets of results need not produce identical time-series patterns.

In Europe (Panel A), internal border cities show an initial relative decline in 1994, followed by a rebound that produces positive deviations from the mid-1990s onward. External border cities follow a similar but delayed trajectory, with coefficients near zero through the early 2000s before turning consistently positive after 2005. At first glance, these positive deviations appear to contradict the population results, which show sustained negative effects. However, the two findings are not necessarily inconsistent. The population results capture relative *growth rates*: border cities grew more slowly than interior cities in every post-1991 period. The nightlights results capture relative *levels* of economic activity measured from a 1992 baseline that is itself already depressed by the initial collapse. The positive nightlights deviations after 2002 may reflect catch-up in economic activity per capita as border city populations shrank while economic output partially stabilized, or they may reflect the normalization of border institutions, EU accession of neighboring countries, and increased cross-border cooperation. The nightlights evidence is therefore best interpreted as showing that the acute economic disruption at borders was concentrated in the 1990s, with some recovery in relative economic activity thereafter, even as population continued to decline in relative terms.

In Central Asia (Panel B), internal border cities show positive and increasing deviations through the late 1990s, consistent with initially limited disruption from the enforcement of inter-republican borders. However, these positive trajectories flatten and turn negative from the mid-2000s onward, potentially reflecting increasing state control, higher tariffs, and tighter border enforcement in countries such as Uzbekistan and Turkmenistan. External border cities present a pattern that contrasts sharply with their European counterparts. Whereas European external border cities showed near-zero or negative deviations through the early 2000s, Central Asian external border cities display positive coefficients from the mid-1990s through the early 2000s, with the 2002 coefficient marginally significant. This divergence is consistent with the core finding that the same geopolitical rupture generated opposite market access shocks across regions: European external border cities faced contracting, westward-reorienting economies across the frontier while Central Asian external border cities gained access to previously sealed frontiers. The Central Asian external border gains attenuate and turn negative from 2009 onward, coinciding with the formalization of border regimes and reduced scope for informal shuttle trade. The imprecision of many individual-year coefficients reflects the small sample of external border cities (57 clusters) and the inherent noisiness of DMSP data in sparsely populated regions.

Overall, the nightlights evidence provides partial corroboration of the population findings. The cross-regional contrast in external border trajectories, negative in Europe and positive in Central Asia during the first post-collapse decade, mirrors the opposing signs documented in the population regressions and reinforces the market access interpretation. The two outcome measures capture different dimensions of the border shock, however, and should not be expected to produce identical time-series patterns.